

EE104 Fundamentals of Electric Circuits (Lesson 11)

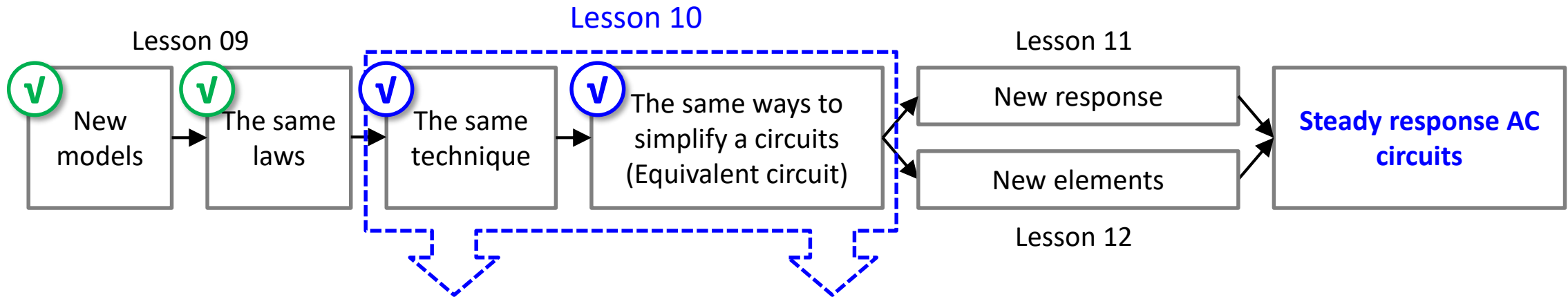
Jun MA | 马俊

E-mail: maj3@sustech.edu.cn

南方科技大学-工学院-电子与电气工程系-PN Lab

Review of Lesson 10 (a)

- EE104 road map



Techniques and theorems

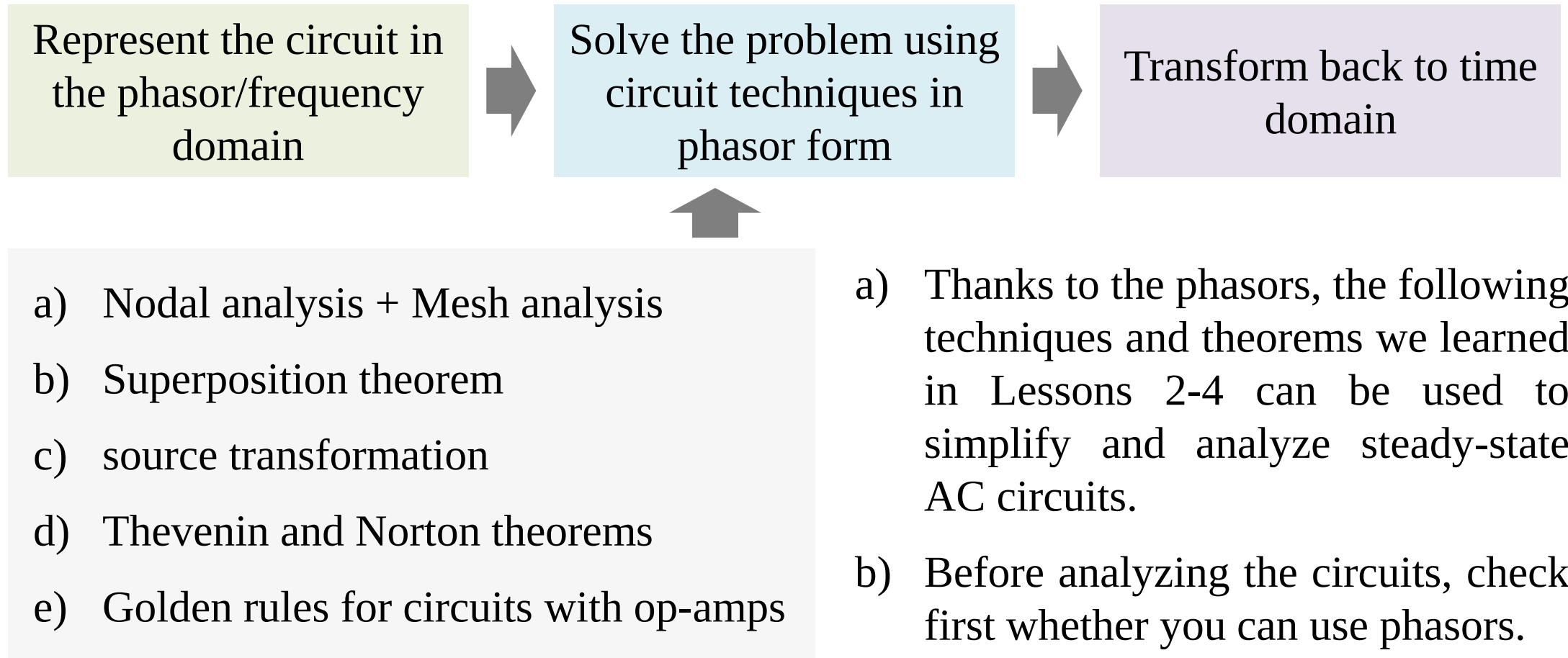
- Nodal analysis and mesh analysis
- Superposition and source transformation
- Thevenin and Norton theorems
- AC op amp circuits

Power

- Power in time domain
- Power in phasor domain (complex power)
- Power factor
- Maximum power transfer theorem

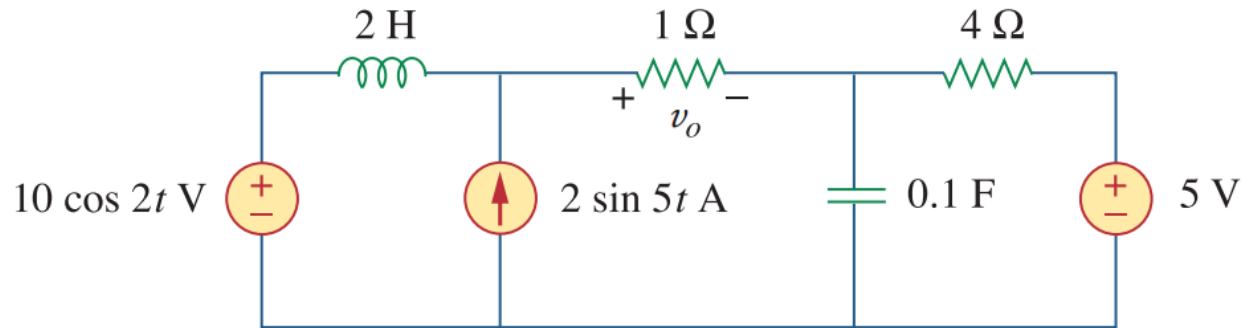
Review of Lessons 10(a)

- Procedures to analyze a steady-state AC circuits:

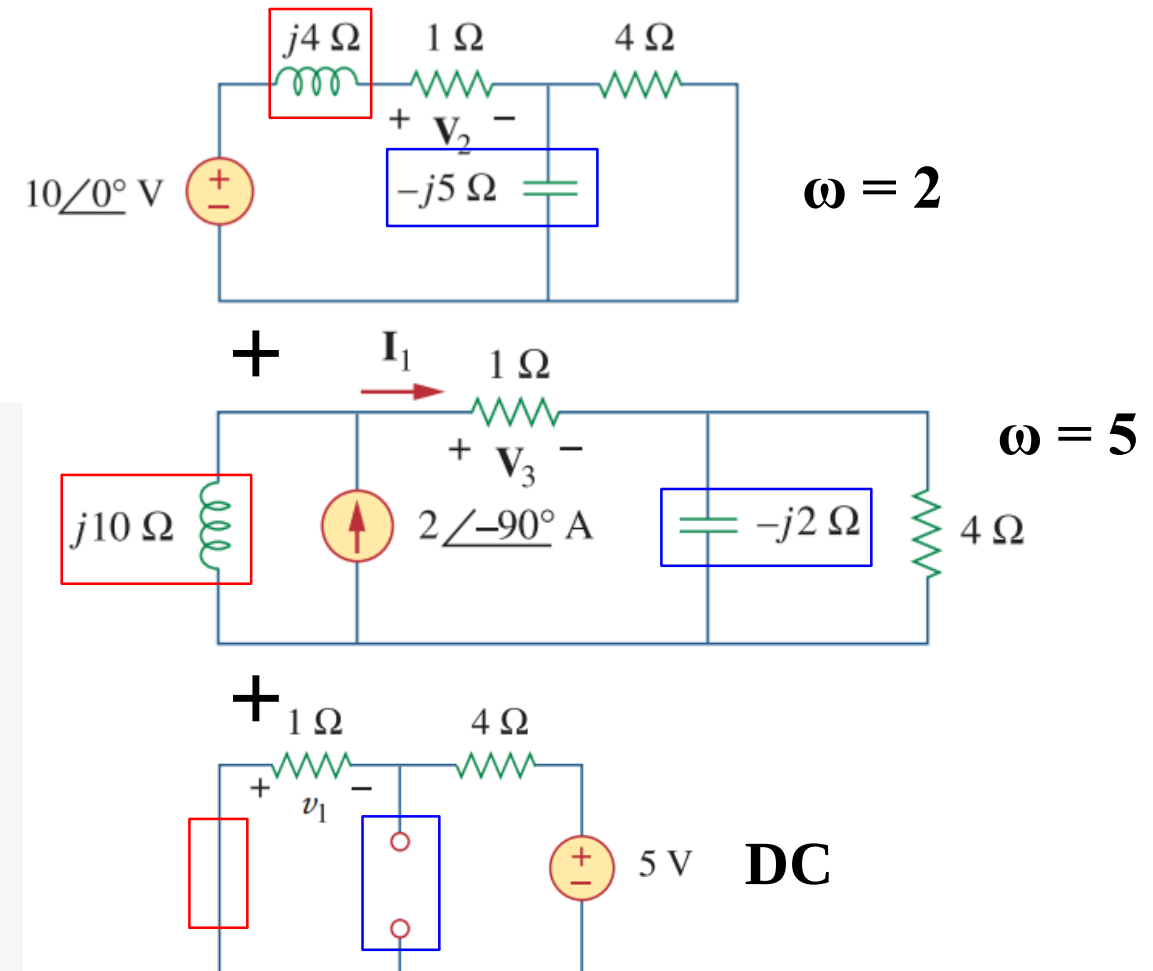


Review of Lessons 10(b)

- Superposition theorem for circuits with different frequencies:



- Keep in mind that the impedance can vary with the frequency when using superposition theorem to analyze an AC circuit.
- For steady-state DC circuits, the inductor is like short circuit while a capacitor is like open circuit.



Review of Lessons 10(c)

- AC power in time domain:

Instantaneous power
瞬时功率

$$p(t) = \boxed{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)} + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Average power
平均功率

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Effective values
有效值

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}, \quad I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

Power factor | 功率因数

$$\text{pf} = P/S = \cos(\theta_v - \theta_i)$$

Apparent power | 视在功率

$$S = V_{\text{rms}} I_{\text{rms}}$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i$$

Review of Lessons 10(d)

- AC power in phasor domain:

$$\text{Complex Power} = \mathbf{S} = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^*$$

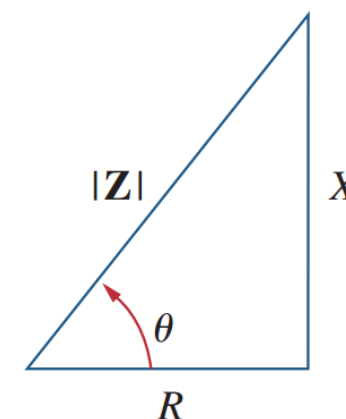
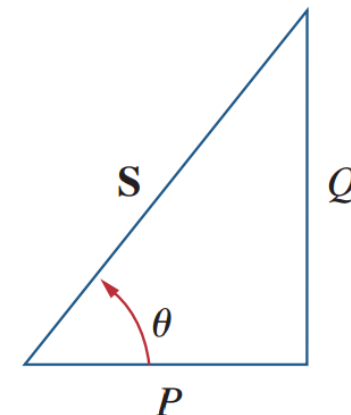
$$= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \angle \theta_v - \theta_i$$

$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$$

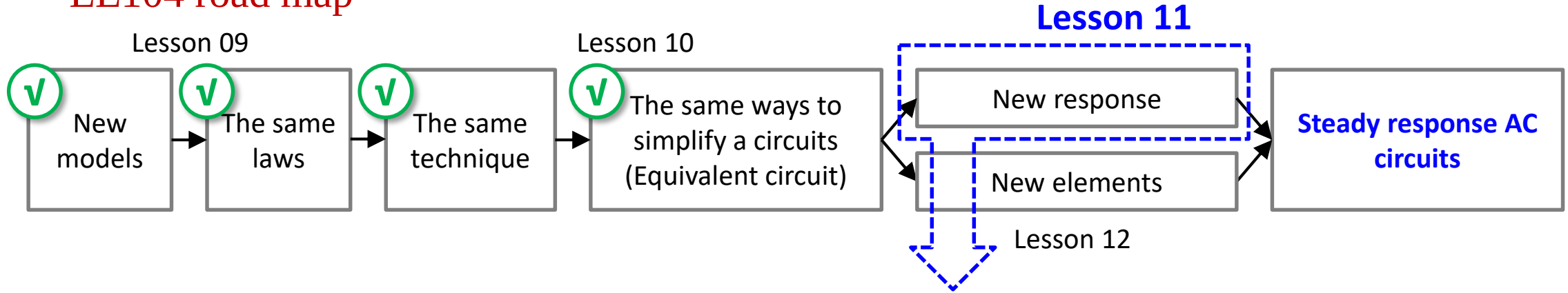
$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$



Review of Lesson 10 (a)

- EE104 road map



Frequency response

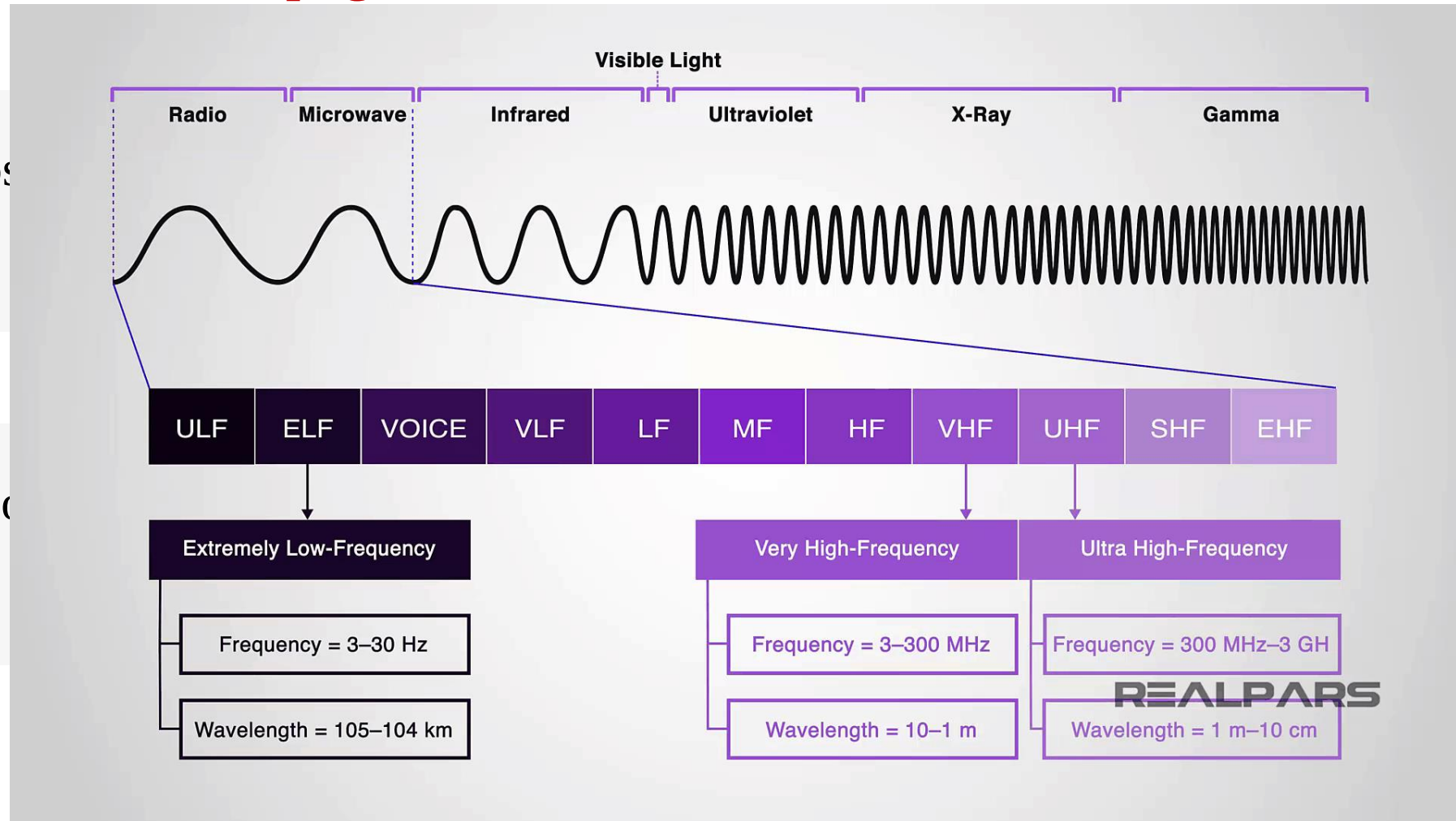
- Transfer function
- Passive filters: lowpass and highpass filters
- Resonance + bandpass and bandstop filters
- Active filters
- High-order filter

Lesson 11 Frequency response

11.1-11.4	Transfer function, The Decibel Scale, Bode Plots	Slides 09-20
11.5-11.6	Lowpass and highpass filters	Slides 22-37
11.7-11.8	Series and parallel resonances	Slides 39-54
11.9	Bandpass and bandstop filters	Slides 56-66
11.10	Active filters	Slides 68-75
11.11	High-order filters and scaling	Slides 77-81
11.12	Summary	Slides 83-85

11.1 Frequency response (a)

- On textbook page 613:



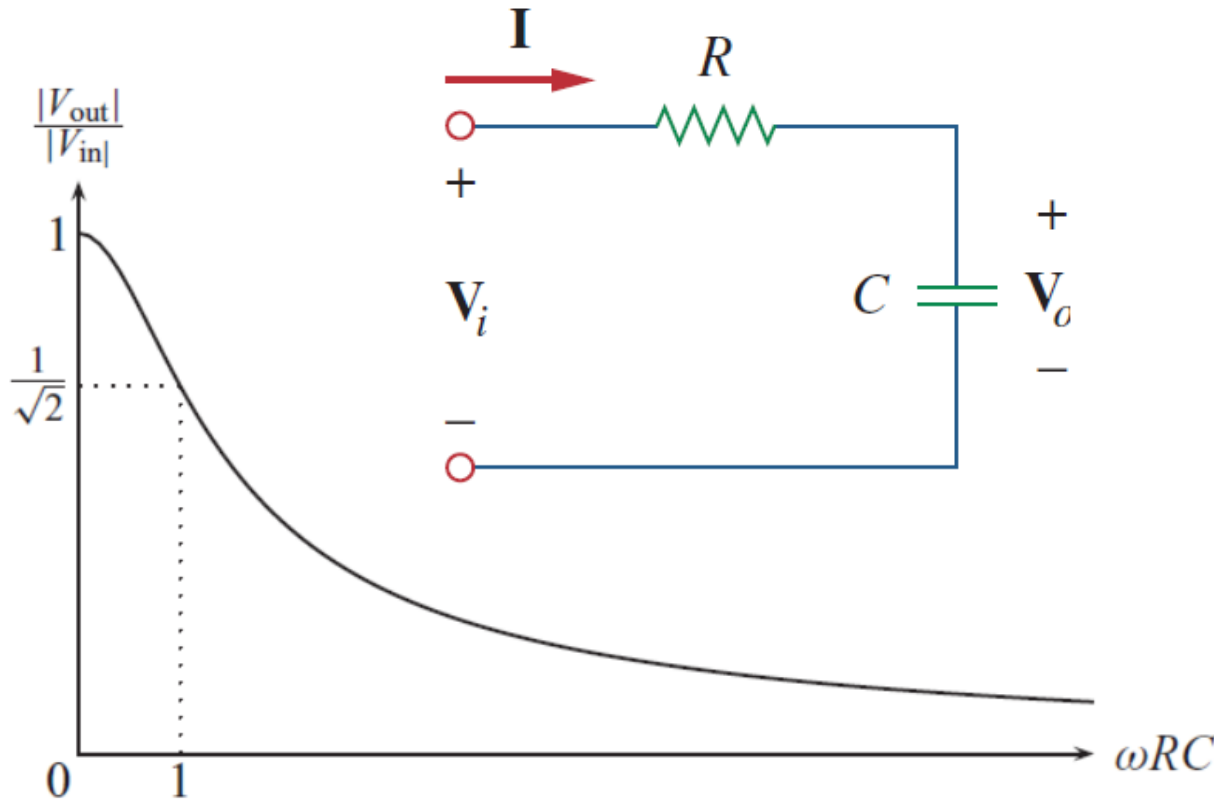
11.1 Frequency response (b)

- Frequency response:

- a) We have learned how to calculate the **steady-state response** of a circuit to AC signals/sources using the technique of phasor representation. Therein, **the signal frequency is assumed to be constant and the impedance of elements are fixed**.
- b) In this lesson, we are going to learn the behavior of a circuit to source of **varying frequency**.
- c) **Frequency response** is the quantitative measure of the output spectrum of a circuit or device in response to a stimulus, and is used to characterize the dynamics of the system. It is a measure of magnitude and phase of the output as a function of frequency, in response to the input.

11.1 Frequency response (c)

- Look at this negative phase shifter:



- a) If the values of R and C are fixed, it is clear that response in amplitude, i.e. $|V_o|/|V_i|$, decreases as the ω increase as:

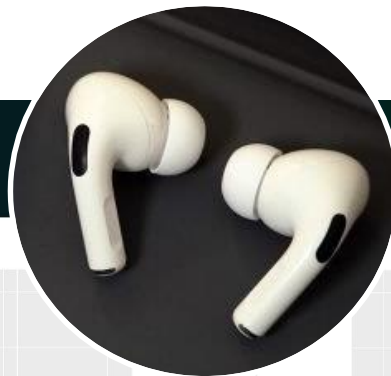
$$\frac{|V_o|}{|V_i|} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

- a) The changing ω also impacts the phase shift as

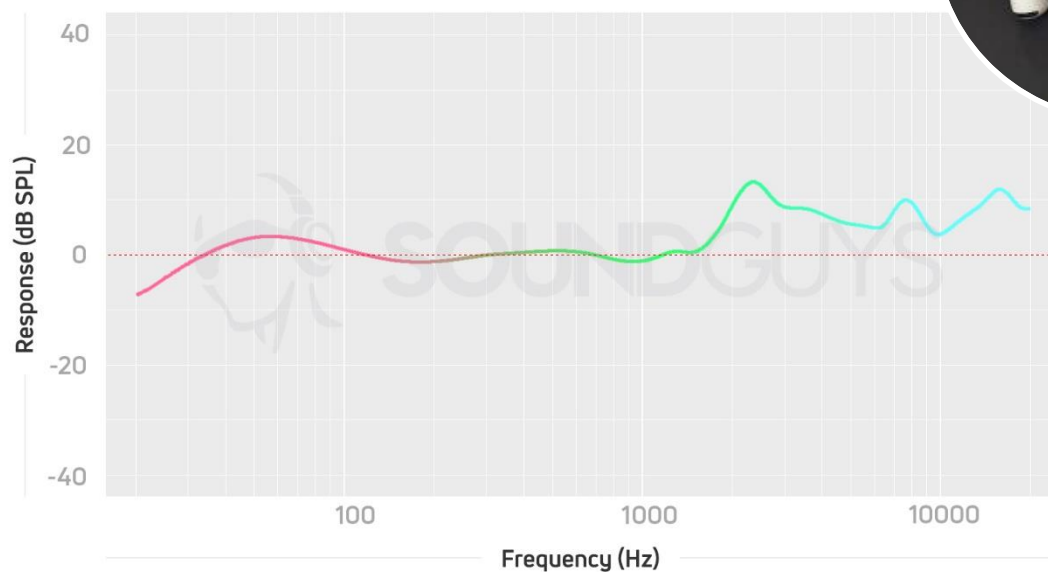
$$\theta = -\tan^{-1}(\omega RC)$$

11.1 Frequency response (d)

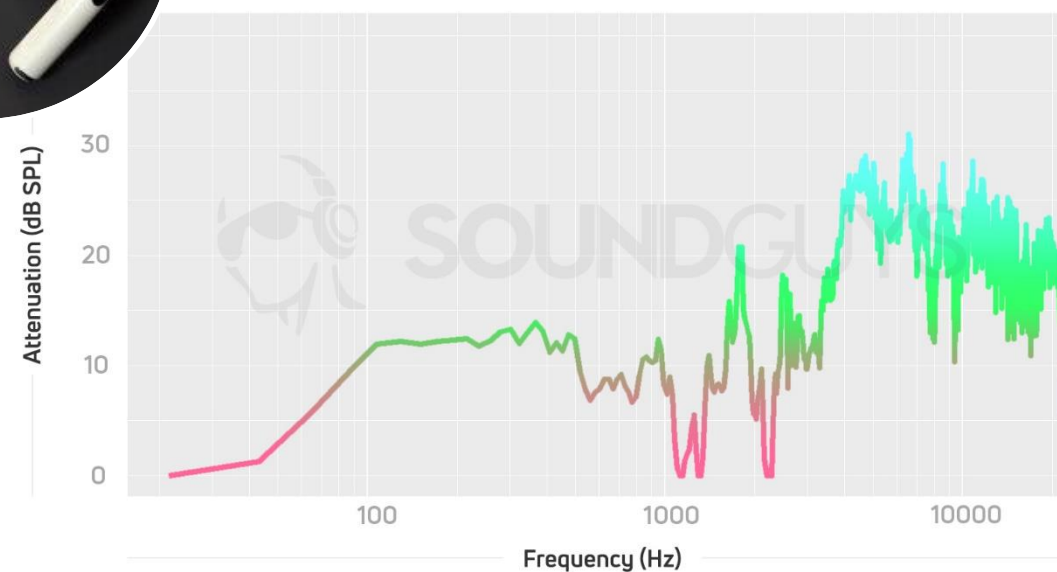
- Why is frequency response important:



Apple AirPods Pro Frequency Response

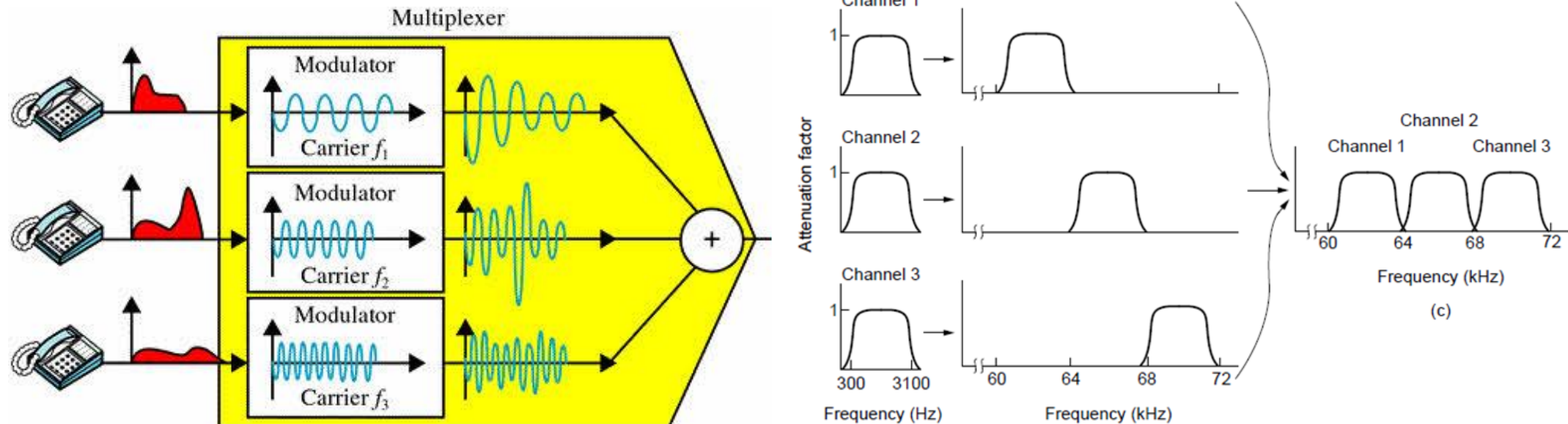


Apple AirPods Pro Isolation



11.1 Frequency response (e)

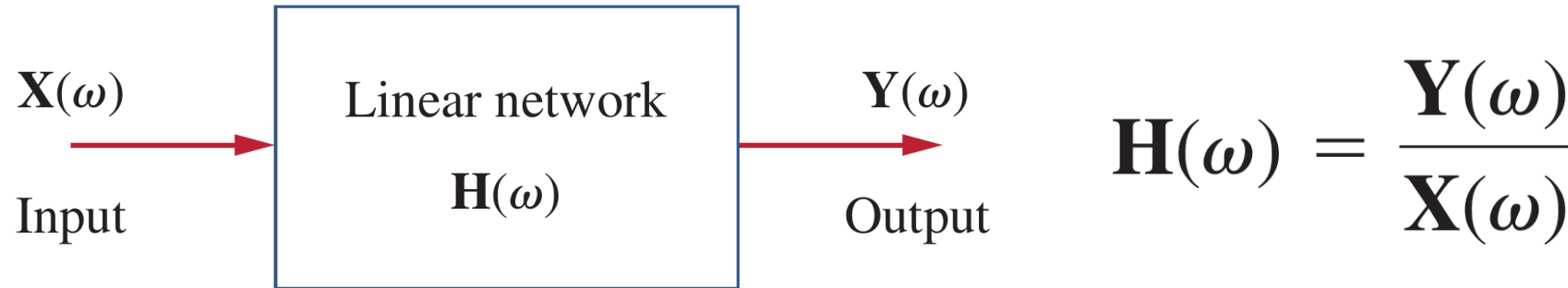
- Why is frequency response important:



In telecommunications, **frequency-division multiplexing (FDM, 频分多路复用)** is a technique by which the total bandwidth available in a communication medium is divided into a series of non-overlapping frequency bands, each of which is used to carry a separate signal. This allows a single transmission medium such as a cable or optical fiber to be shared by multiple independent signals.

11.2 Transfer function (a)

- The transfer function (TF, 传递函数):



- The transfer function $\mathbf{H}(\omega)$ of a circuit is the frequency-dependent ratio of a phasor output $\mathbf{Y}(\omega)$ (an element voltage or current) to a phasor input $\mathbf{X}(\omega)$ (source voltage or current).
- The transfer function $\mathbf{H}(\omega)$ (also called the **network function**) is a useful analytical tool for finding the frequency response of a circuit. In fact, the frequency response of a circuit is the plot of the circuit's transfer function $\mathbf{H}(\omega)$ versus ω , with ω varying from 0 to ∞ .

11.2 Transfer function (b)

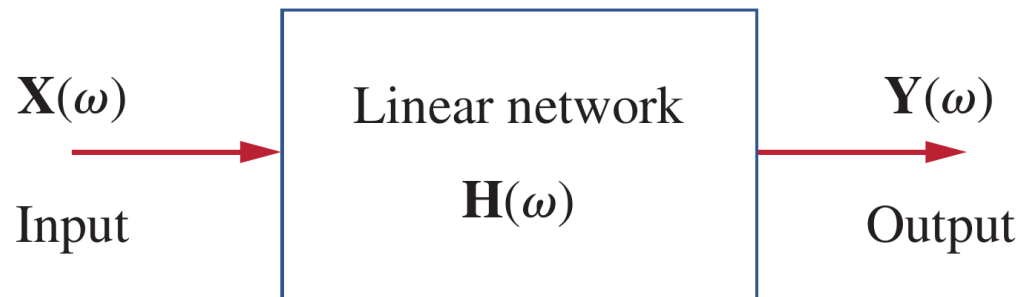
- The transfer function (TF, 传递函数):

a) Since the input and output can be either voltage or current at any place in the circuit, there are four possible transfer functions.

Input: voltage	Output: voltage	$\mathbf{H}(\omega) = \text{Voltage gain} = \frac{\mathbf{V}_o(\omega)}{\mathbf{V}_i(\omega)}$	电压增益
Input: current	Output: current	$\mathbf{H}(\omega) = \text{Current gain} = \frac{\mathbf{I}_o(\omega)}{\mathbf{I}_i(\omega)}$	电流增益
Input: current	Output: voltage	$\mathbf{H}(\omega) = \text{Transfer Impedance} = \frac{\mathbf{V}_o(\omega)}{\mathbf{I}_i(\omega)}$	转移阻抗
Input: voltage	Output: current	$\mathbf{H}(\omega) = \text{Transfer Admittance} = \frac{\mathbf{I}_o(\omega)}{\mathbf{V}_i(\omega)}$	转移导纳

11.2 Transfer function (c)

- The transfer function (TF, 传递函数):



$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)} = |H(\omega)| e^{j\varphi(x)}$$

Magnitude response
幅频响应

phase response
相频响应

$$\begin{aligned} \mathbf{Y}(\omega) &= \mathbf{H}(\omega)\mathbf{X}(\omega) \\ &= |\mathbf{H}(\omega)| |\mathbf{X}(\omega)| \underline{\theta_x + \theta_H} \end{aligned}$$

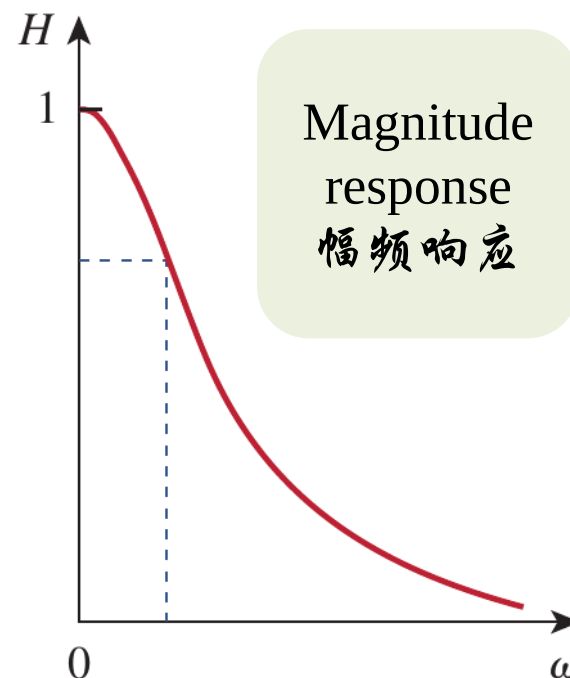
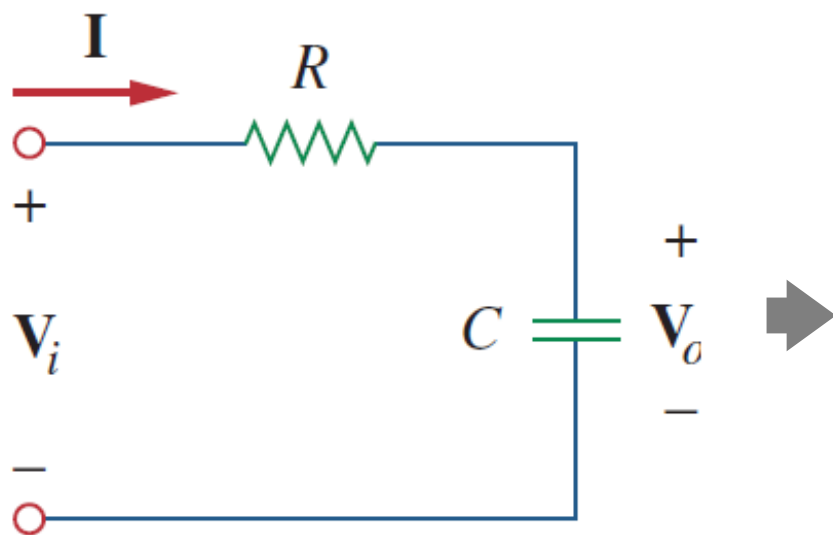
11.2 Transfer function (d)

- An example: TF of negative phase shifters:

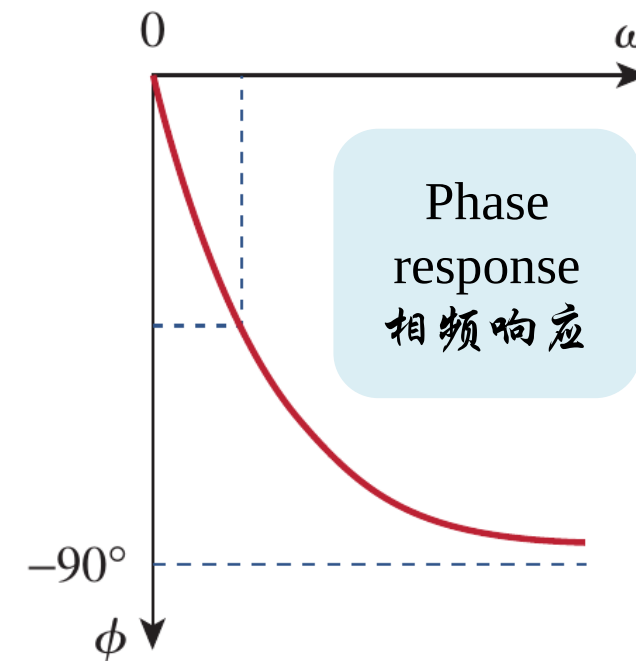
$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{1}{1 + j\omega RC}$$



$$H = \frac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$



$$\Delta\theta = -\tan^{-1}(\omega RC)$$



11.3 The Decibel Scale (a)

- The Decibel Scale (分贝表示法):

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1}$$

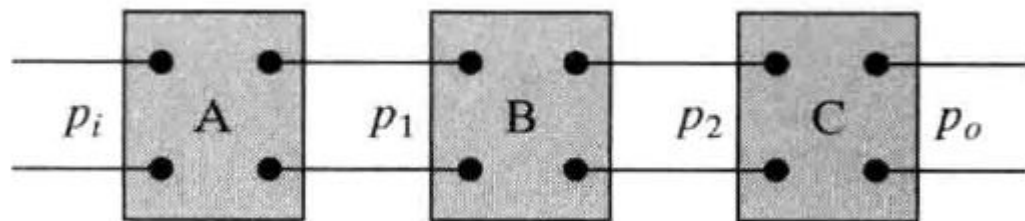
$$G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} \quad G_{\text{dB}} = 20 \log_{10} \frac{I_2}{I_1}$$

The **decibel** (symbol: **dB**) is a unit of measurement used to express the ratio between two values of a physical property on a logarithmic scale. It can be used to express a relative change or an absolute value related to a reference. For the latter, the symbol may change to reflect the unit and property of the reference, such as “dBV” or “dBm” when the reference is 1 V or 1 mW.

dB	Power ratio	Amplitude ratio
60	1e6 (一百万)	1000
30	1000	31.62
10	10	3.162
6	3.981 $\approx$ 4	1.995 $\approx$ 2
3	1.995 $\approx$ 2	1.413 $\approx \sqrt{2}$
1	1.259	1.122
0	1	1
-3	0.5	0.708
-6	0.251	0.5

11.3 The Decibel Scale (b)

- The Decibel Scale (分贝表示法):



$$\left. \begin{aligned} \sigma_A &= \frac{p_1}{p_i} & \sigma_B &= \frac{p_2}{p_1} & \sigma_C &= \frac{p_o}{p_2} \\ \sigma_{\text{eff}} &= \frac{p_o}{p_i} = \frac{p_1}{p_i} \frac{p_2}{p_1} \frac{p_o}{p_2} = \sigma_A \sigma_B \sigma_C \end{aligned} \right\}$$

$$\left. \begin{aligned} \sigma_{\text{eff}} &= \frac{p_o}{p_i} = \frac{v_o^2 / R_L}{v_i^2 / R_i} = \left(\frac{v_o}{v_i} \right)^2 \\ \sigma_{\text{eff}} &= \frac{p_o}{p_i} = \frac{i_o^2 R_L}{i_i^2 R_i} = \left(\frac{i_o}{i_i} \right)^2 \end{aligned} \right\}$$

$$\lg(\sigma_{\text{eff}}) = \lg(\sigma_A) + \lg(\sigma_B) + \lg(\sigma_C)$$

$$\text{define, } G_{db} = 10 \lg \left(\frac{p_o}{p_i} \right)$$

$$G_{db} = 20 \lg \left(\frac{v_o}{v_i} \right) \text{ and } G_{db} = 20 \lg \left(\frac{i_o}{i_i} \right)$$

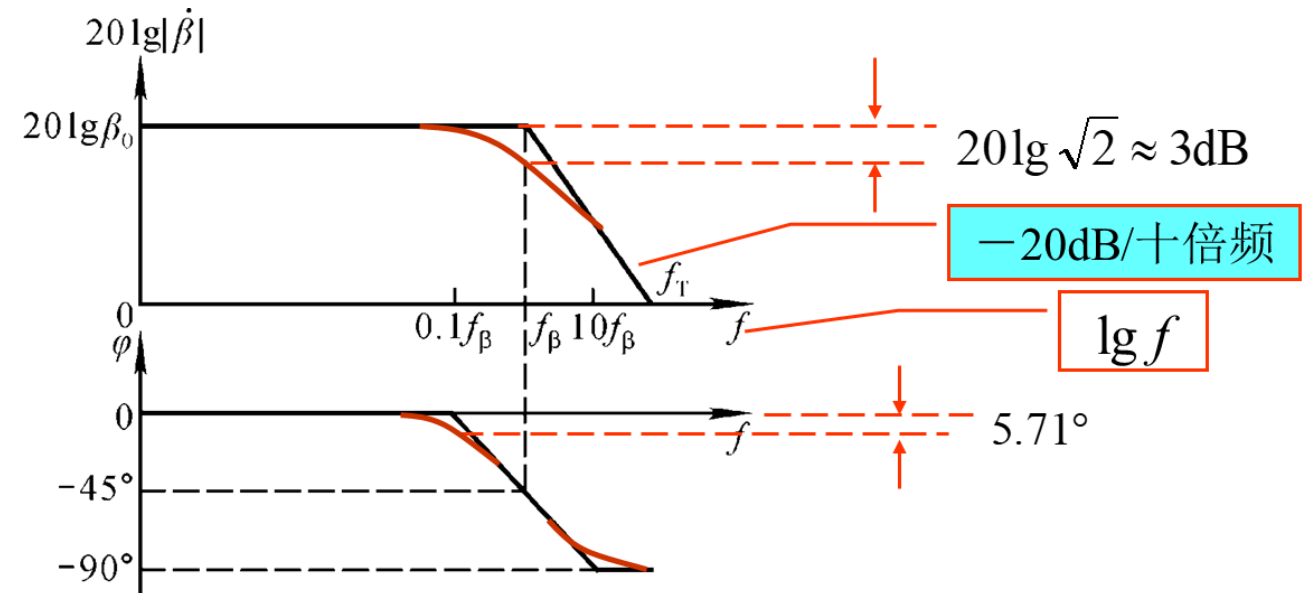
11.4 Bode plots (a)

- Bode plots (伯德图) of circuits:

$$\mathbf{H} = H/\phi = He^{j\phi}$$

$$\ln \mathbf{H} = \ln H + \ln e^{j\phi} = \ln H + j\phi$$

$$H_{\text{dB}} = 20 \log_{10} H$$



- a) A Bode plot is a graph of the frequency response of a system. It is usually a combination of a Bode magnitude plot, expressing the magnitude response (usually in decibels), and a phase response. The use of log scaling in both x and y makes it possible to use polygonal lines to approximate the actual curves, to cover broad range of values and to ease operations.

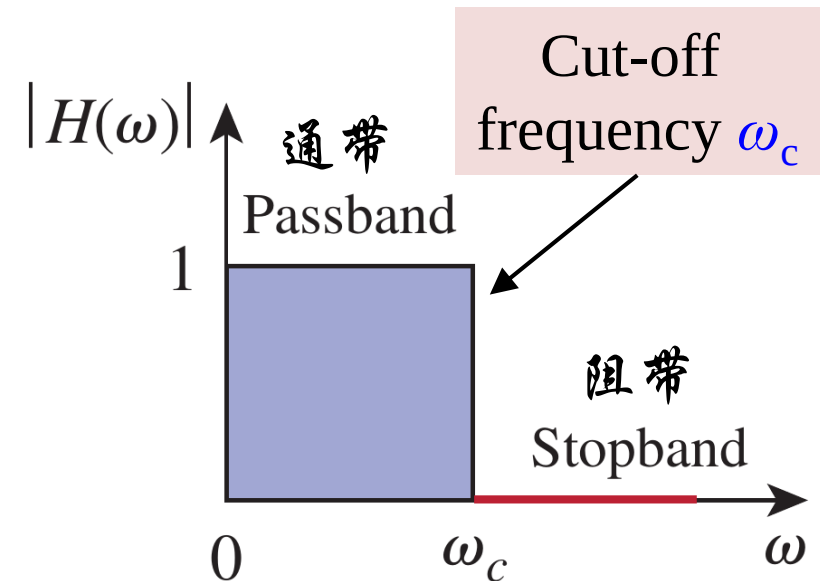
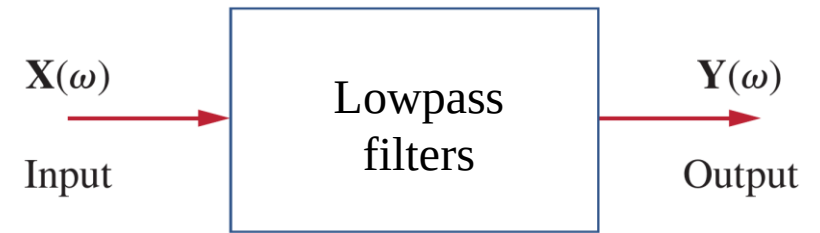
Lesson 11 Frequency response

11.1-11.4	Transfer function, The Decibel Scale, Bode Plots	Slides 09-20
11.5-11.6	Lowpass and highpass filters	Slides 22-37
11.7-11.8	Series and parallel resonances	Slides 39-54
11.9	Bandpass and bandstop filters	Slides 56-66
11.10	Active filters	Slides 68-75
11.11	High-order filters and scaling	Slides 77-81
11.12	Summary	Slides 83-85

11.5 Lowpass filters (a)

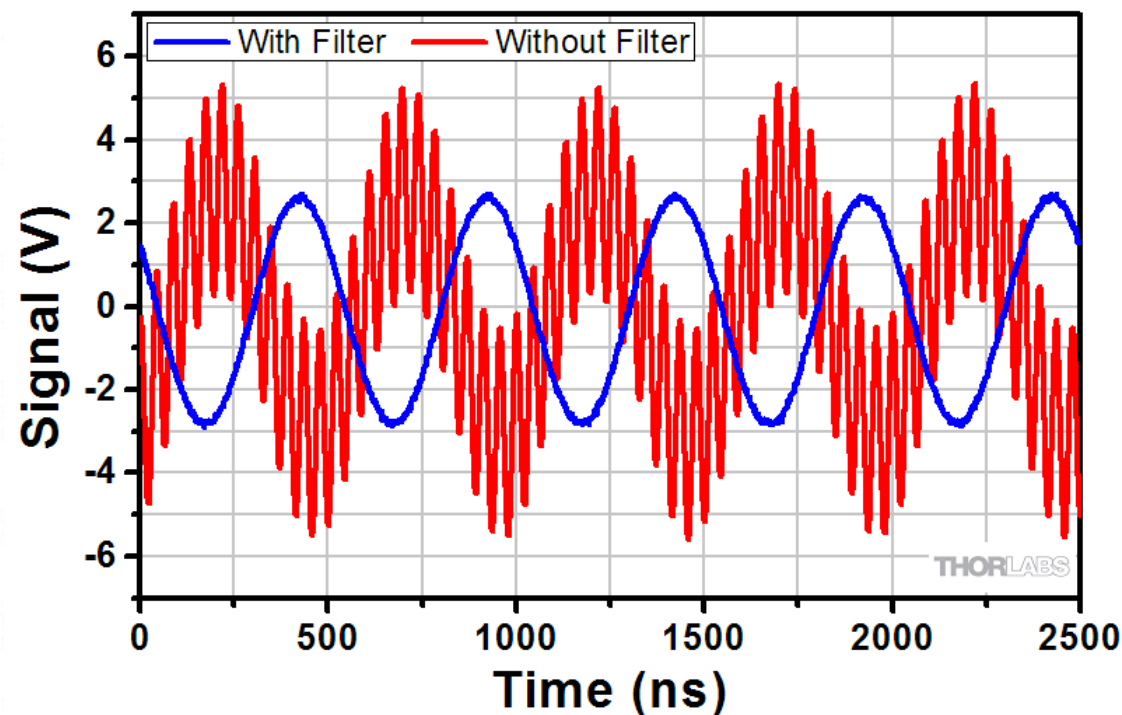
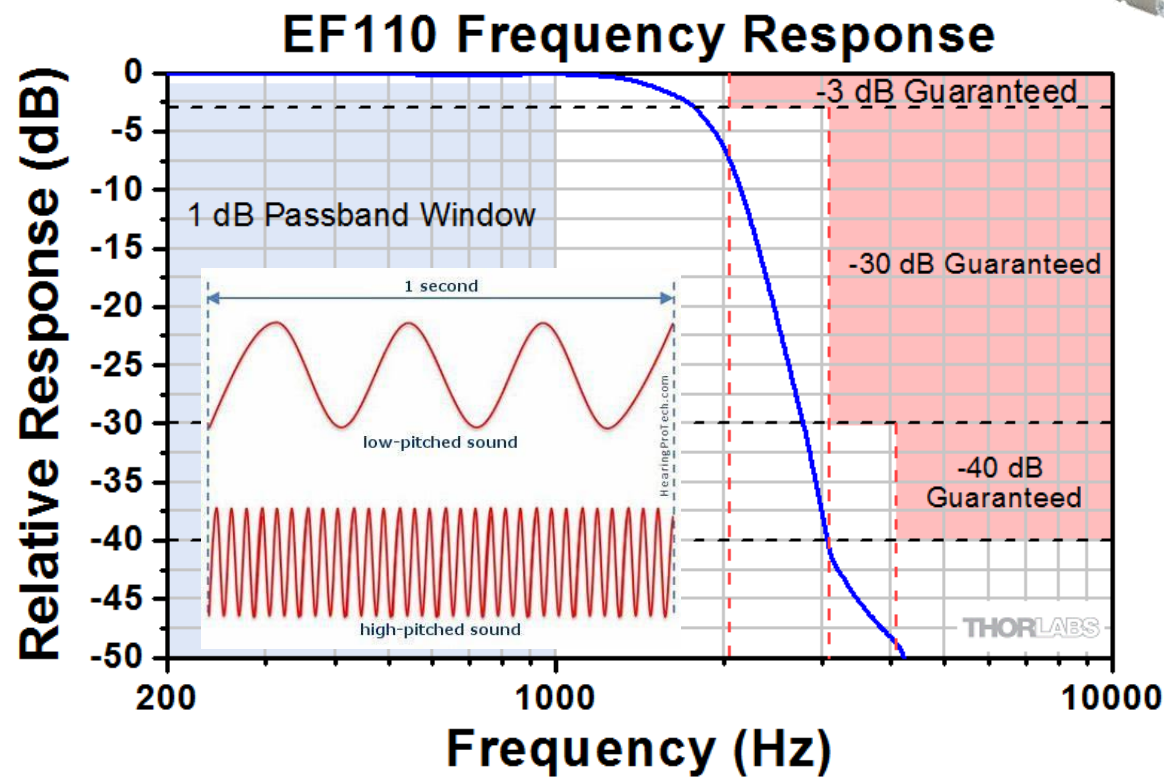
- Filters (滤波器) and ideal lowpass filters (低通滤波器):

- a) A filter is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.
- b) Filters are of great importance and have been an integral part of the evolution of electrical engineering from the beginning. For examples, filters are used in radio and TV receivers to allow us to select one desired signal out of a multitude of broadcast signals in the environment.
- c) A filter is a **passive filter** if it consists of only passive elements R, L, and C. It is said to be an active filter if it consists of active elements (such as transistors and op amps) in addition to passive elements R, L, and C.
- d) A **lowpass filter** passes low frequencies and stops high frequencies



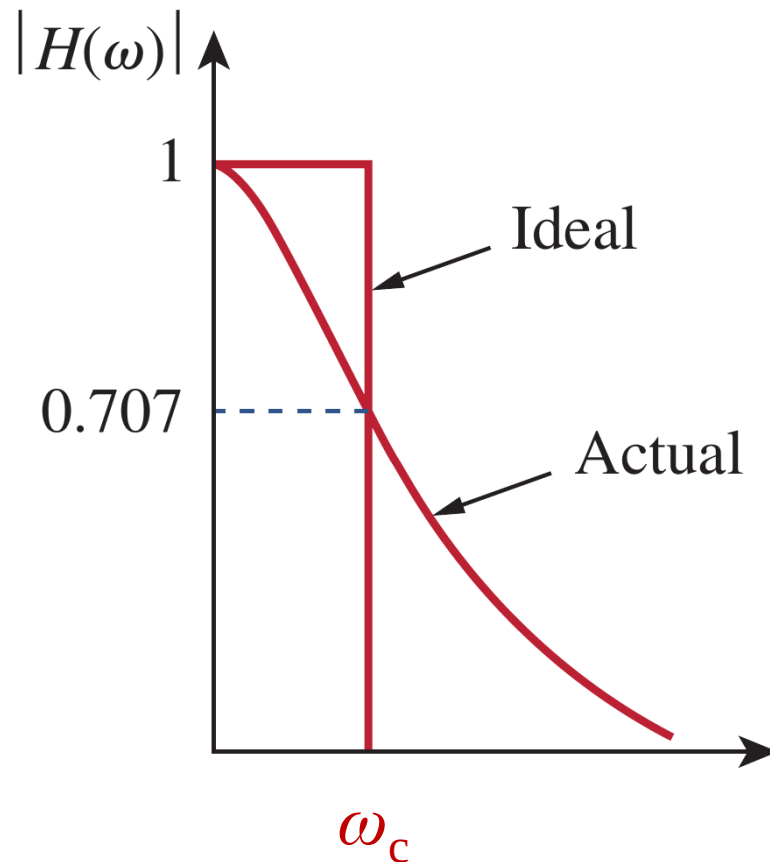
11.5 Lowpass filters (b)

- Practical lowpass filters:

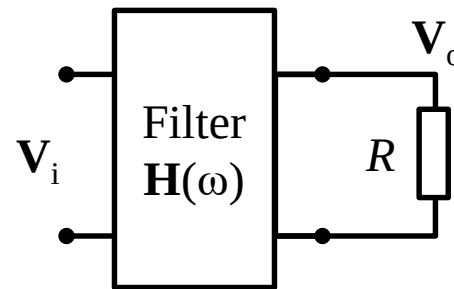


11.5 Lowpass filters (c)

- Cut-off frequencies (ω_c , 截止频率):



- The cutoff frequency is the frequency at which the transfer function **H** drops in magnitude to 70.71% ($1/\sqrt{2}$) of its maximum value.
- The cut-off frequencies is also called rolloff frequency (滚降频率) or **half-power frequency** (半功率点频率).

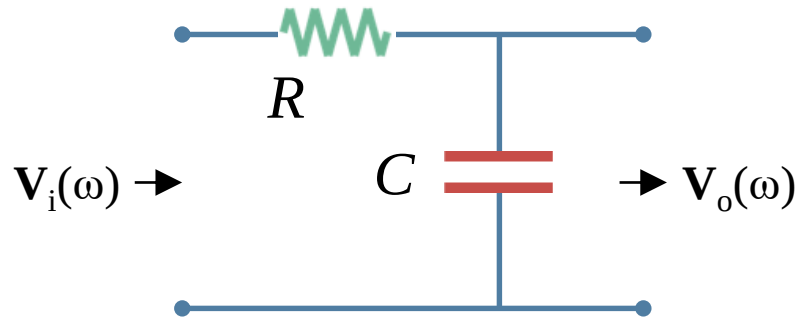


$$\begin{aligned}
 \text{At } \omega_c, |\mathbf{V}_o| &= |\mathbf{H}(\omega_c)| |\mathbf{V}_i| \\
 &= \frac{1}{\sqrt{2}} H_{max} |\mathbf{V}_i| = \frac{1}{\sqrt{2}} V_{o,max} \\
 P(\omega_c) &= \frac{1}{2} \frac{\left(\frac{1}{\sqrt{2}} V_{o,max}\right)^2}{R} = \frac{1}{2} \frac{1}{2} \frac{V_{o,max}^2}{R} = \frac{P_{max}}{2}
 \end{aligned}$$

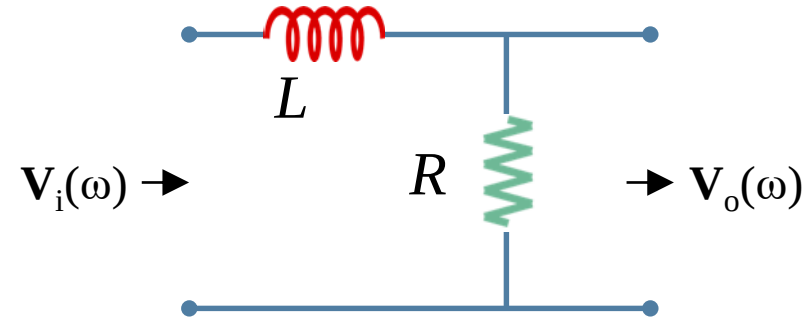
*Here we are talking about average powers.

11.5 Lowpass filters (d)

- Basic lowpass filters:



RC lowpass filters:

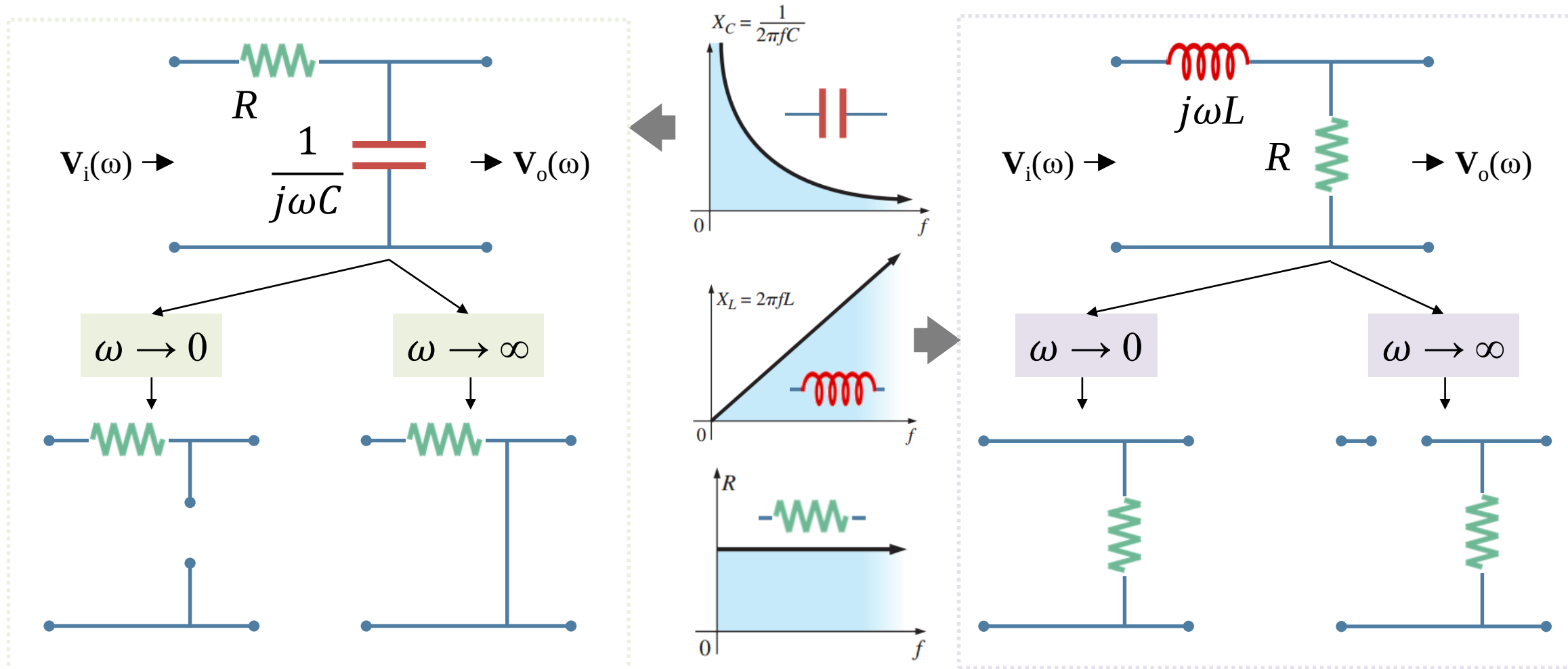


RL lowpass filters:

- a) **Passive lowpass filters** can be based on both RC and RL voltage dividers, which allows only the pass of signals with frequencies lower than the cut-off frequency.
- b) v_o is taken from the capacitor in RC circuits.
- c) v_o is taken from the resistor in RL circuits.
- d) Advance passive lowpass filters can be based on more complicated circuits.

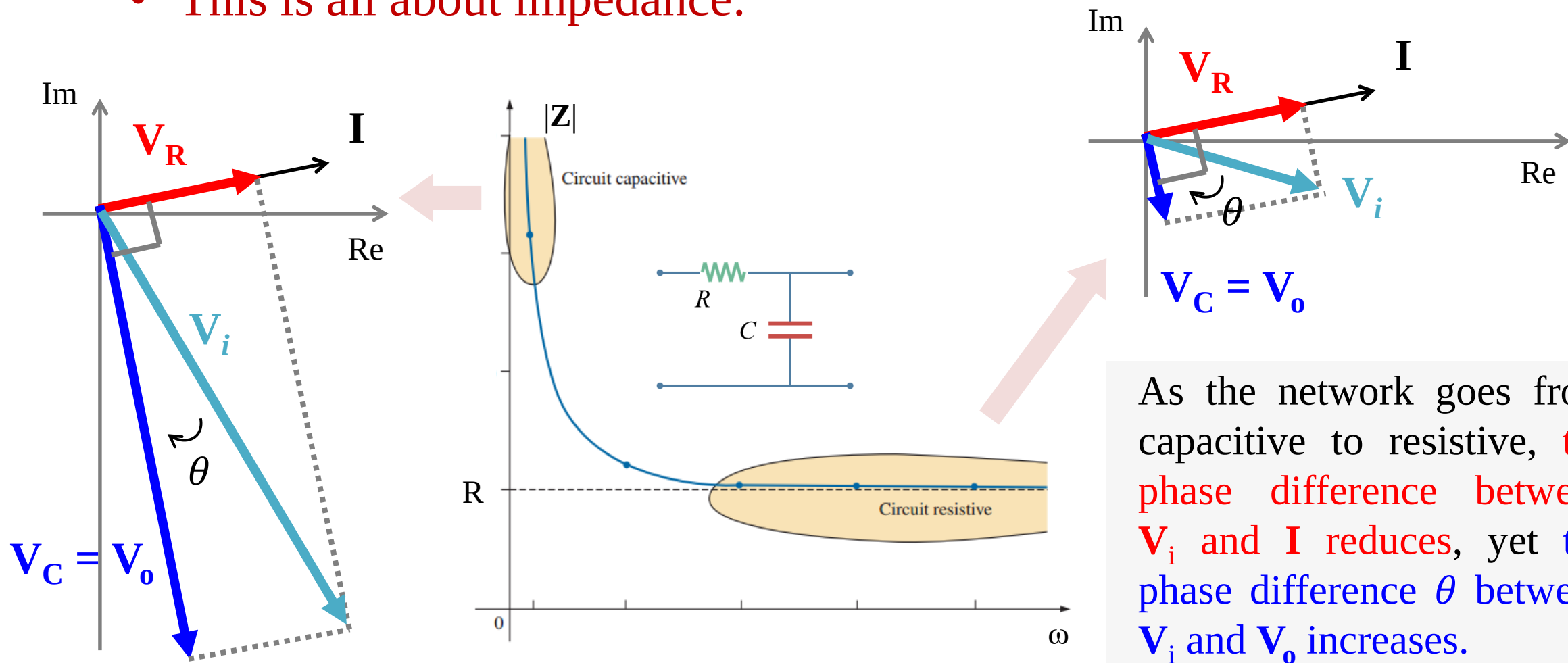
11.5 Lowpass filters (e)

- This is all about impedance:



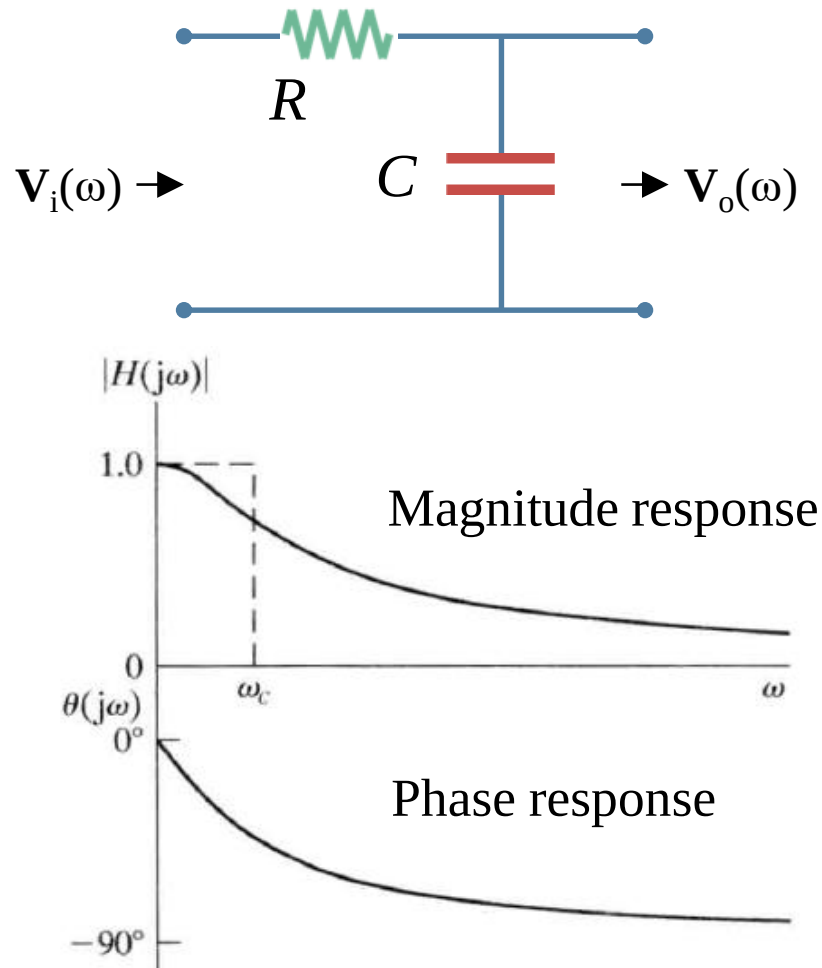
11.5 Lowpass filters (f)

- This is all about impedance:



11.5 Lowpass filters (g)

- RC lowpass filters:



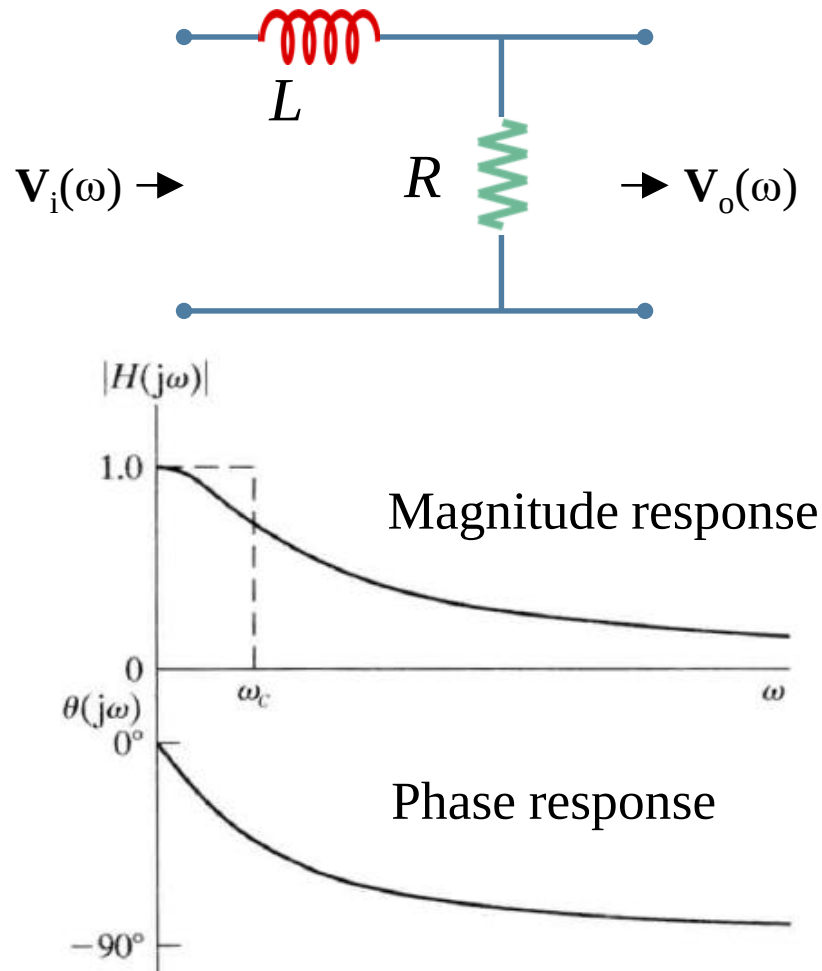
According to voltage division:

$$V_o = \frac{1/j\omega C}{R + 1/j\omega C} V_i \rightarrow \begin{cases} \mathbf{H}(\omega) = \frac{1}{1 + j\omega RC} \\ |\mathbf{H}(\omega)| = \frac{|V_o|}{|V_i|} = \frac{1}{\sqrt{1 + (\omega RC)^2}} \\ \Delta\theta(\omega) = -\tan^{-1}(\omega RC) \end{cases}$$

$$\text{At } \omega_c, |\mathbf{H}(\omega)| = \frac{1}{\sqrt{(\omega RC)^2 + 1}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c = \frac{1}{RC}$$

11.5 Lowpass filters (h)

- RL lowpass filters:



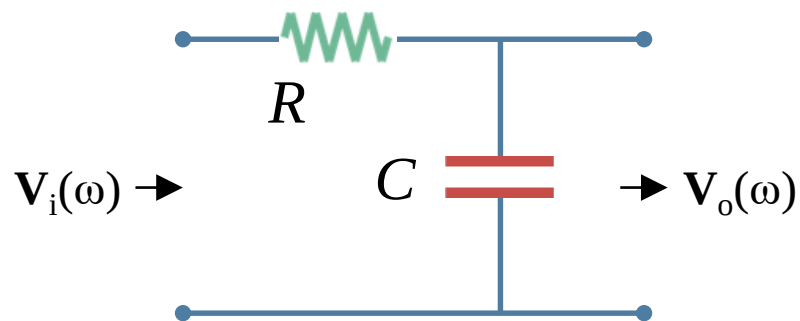
According to voltage division:

$$\mathbf{V}_o = \frac{R}{R + j\omega L} \mathbf{V}_i \rightarrow \begin{cases} \mathbf{H}(\omega) = \frac{R/L}{R/L + j\omega} \\ |\mathbf{H}(\omega)| = \frac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \frac{R/L}{\sqrt{(R/L)^2 + \omega^2}} \\ \Delta\theta(\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right) \end{cases}$$

$$\text{At } \omega_c, |\mathbf{H}(\omega)| = \frac{R/L}{\sqrt{(R/L)^2 + \omega_c^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c = \frac{R}{L}$$

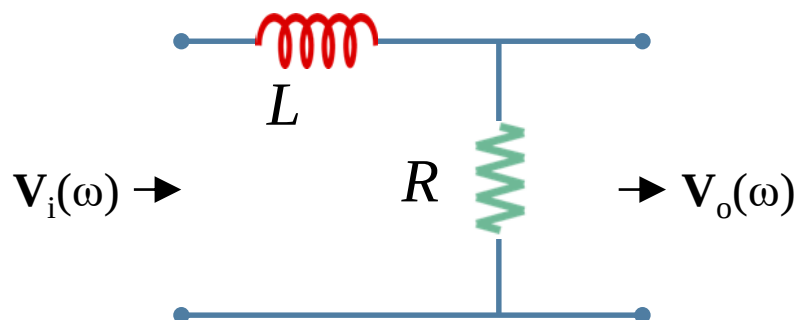
11.5 Lowpass filters (i)

- At the cutoff frequency ω_c :



$$\omega_c = \frac{1}{RC} \Rightarrow \frac{1}{\omega_c C} = R \Rightarrow |X_C| = R$$

What is the phase shift at ω_c ?



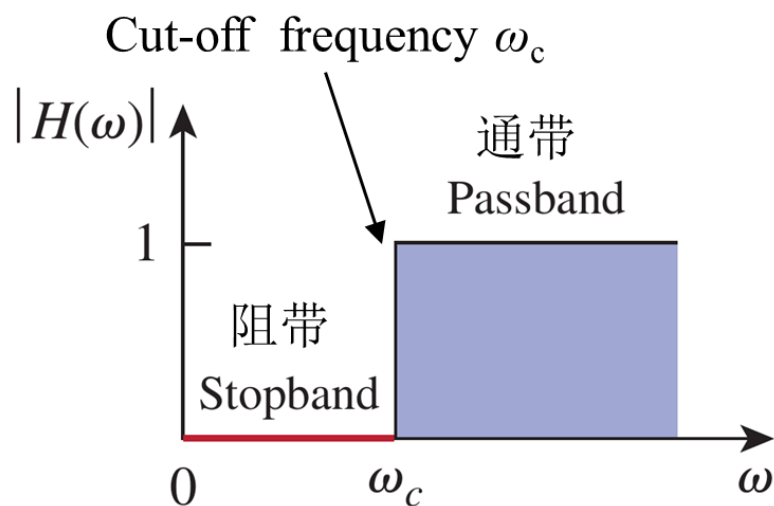
$$\omega_c = \frac{R}{L} \Rightarrow \omega_c L = R \Rightarrow |X_L| = R$$

11.6 Highpass filters (a)

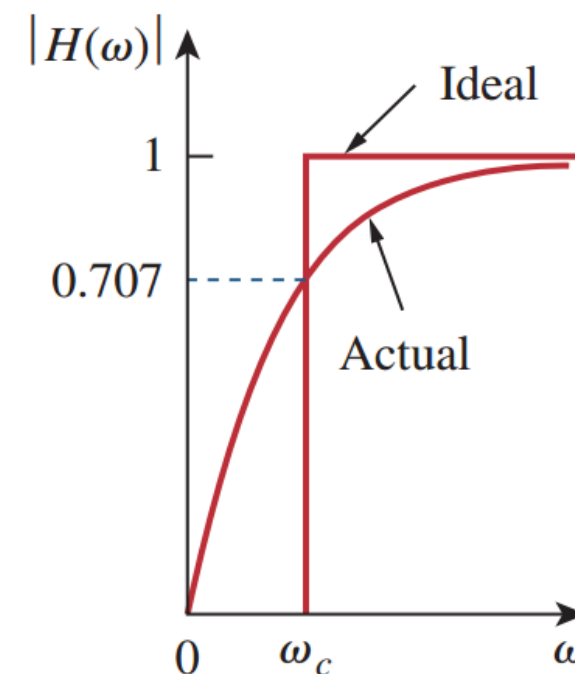
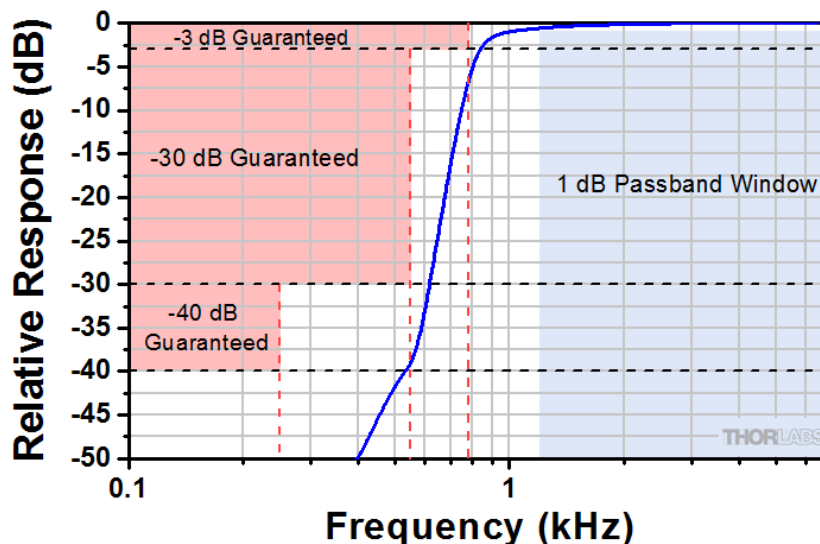
- Highpass filters (高通滤波器):

a) A **highpass filter** is designed to pass all frequencies above its cutoff frequency ω_c .

Ideal highpass filters

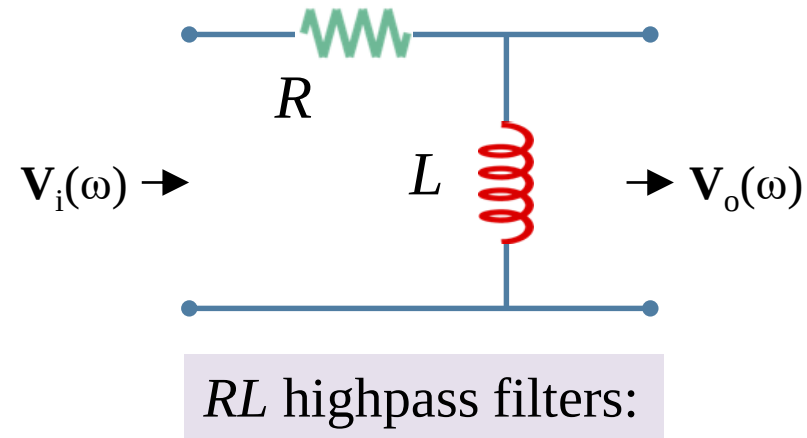
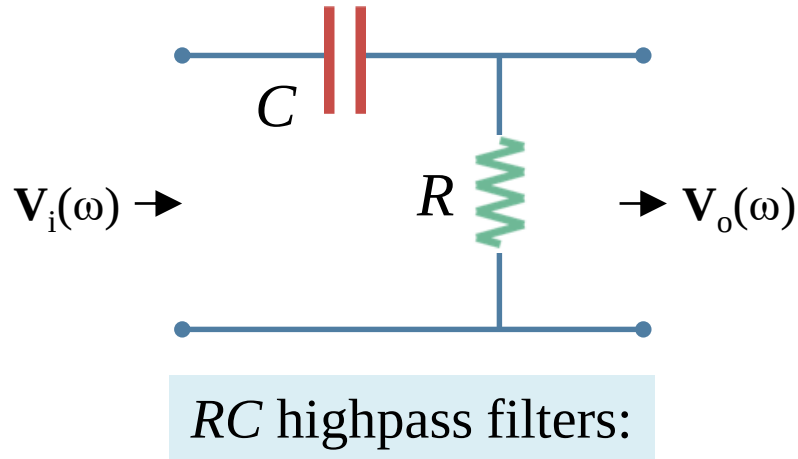


Practical highpass filters



11.6 Highpass filters (b)

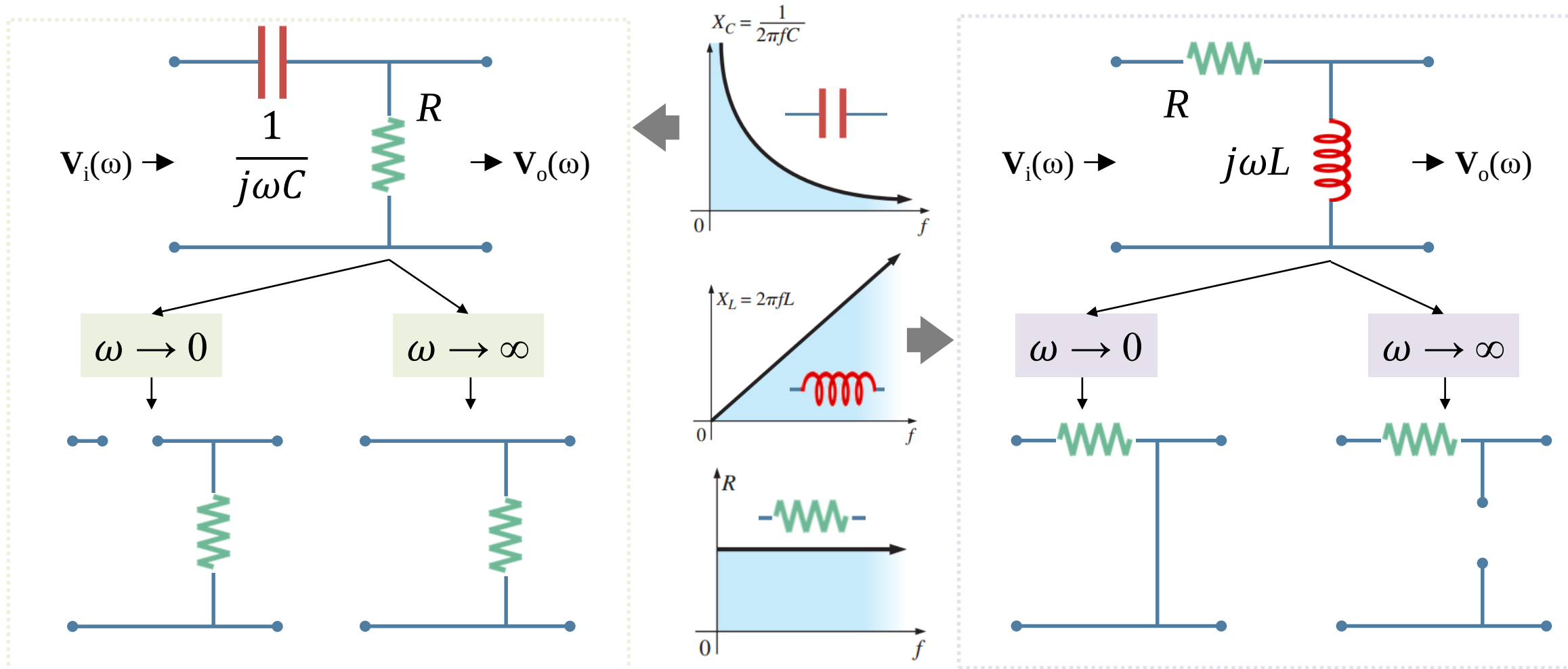
- Basic highpass filters:



- a) **Passive highpass filters** can be based on both *RC* and *RL* voltage dividers, which allows only the pass of signals with frequencies **higher** than the cut-off frequency.
- b) v_o is taken from the resistor in *RC* circuits.
- c) v_o is taken from the inductor in *RL* circuits.
- d) Advance passive highpass filters can be based on more complicated circuits.

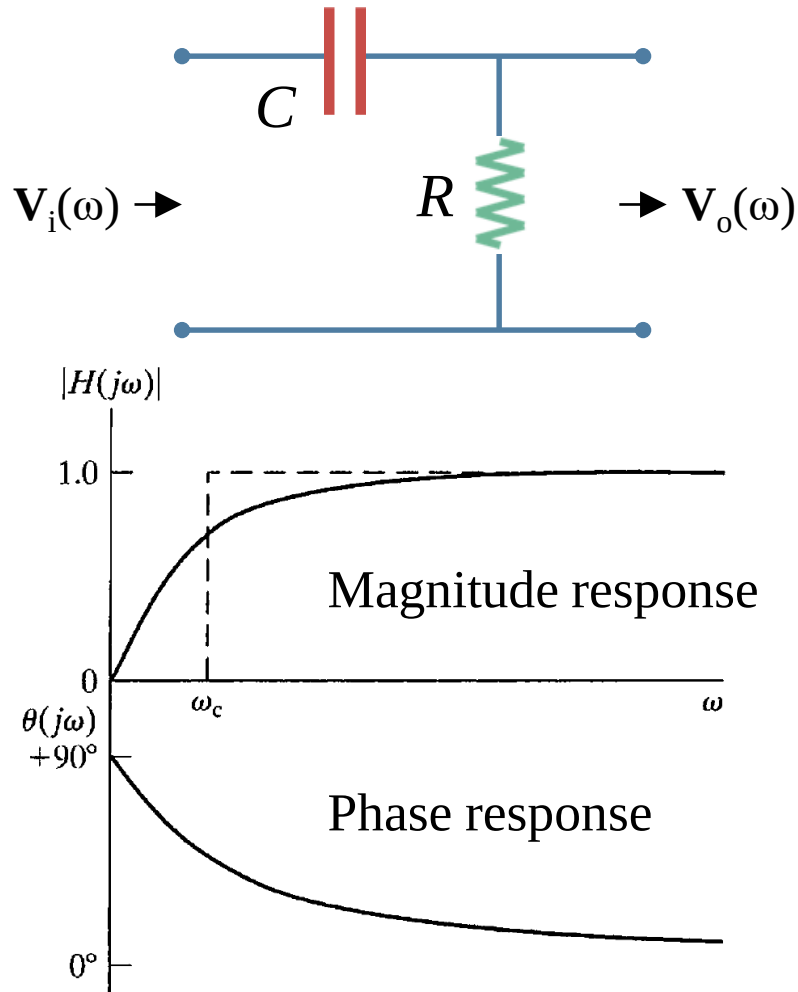
11.6 Highpass filters (c)

- Again this is all about impedance:



11.6 Highpass filters (d)

- RC highpass filters:



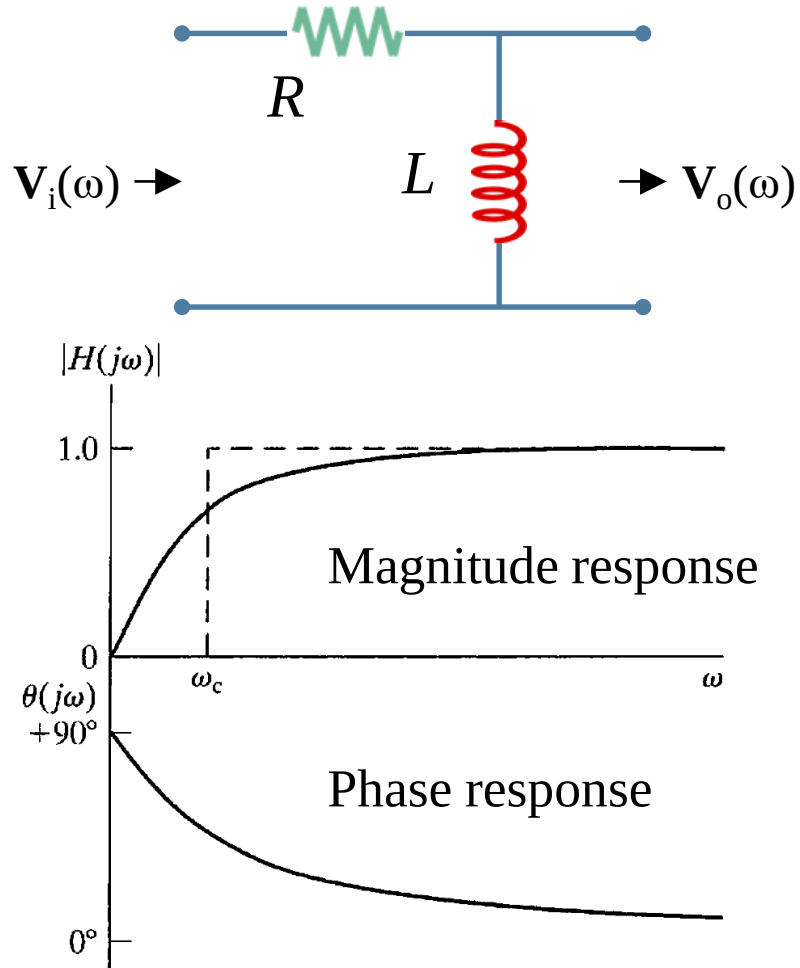
According to voltage division:

$$V_o = \frac{R}{R + 1/j\omega C} V_i \rightarrow \begin{cases} \mathbf{H}(\omega) = \frac{j\omega RC}{1 + j\omega RC} \\ |\mathbf{H}(\omega)| = \frac{|V_o|}{|V_i|} = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}} \\ \Delta\theta(\omega) = \tan^{-1}(1/\omega RC) \end{cases}$$

$$\text{At } \omega_c, |\mathbf{H}(\omega)| = \frac{\omega RC}{\sqrt{(\omega RC)^2 + 1}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c = \frac{1}{RC}$$

11.6 Highpass filters (e)

- RL highpass filters:



According to voltage division:

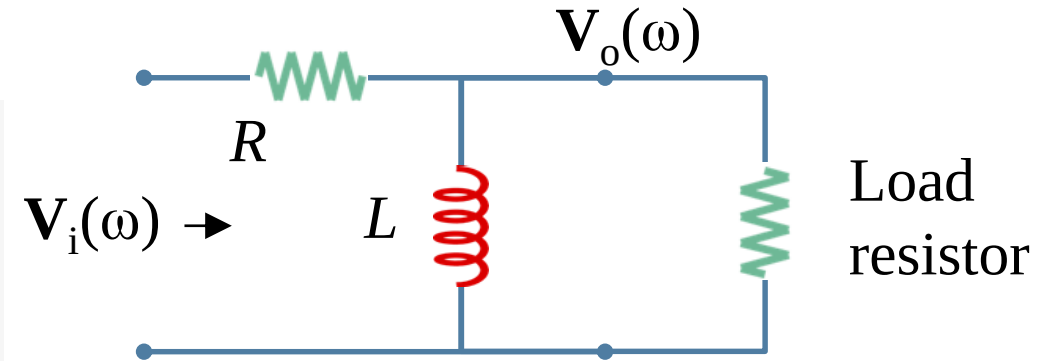
$$\mathbf{V}_o = \frac{j\omega L}{R + j\omega L} \mathbf{V}_i \rightarrow \begin{cases} \mathbf{H}(\omega) = \frac{j\omega}{R/L + j\omega} \\ |\mathbf{H}(\omega)| = \frac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \frac{\omega}{\sqrt{(R/L)^2 + \omega^2}} \\ \Delta\theta(\omega) = \tan^{-1}\left(\frac{R}{\omega L}\right) \end{cases}$$

$$\text{At } \omega_c, |\mathbf{H}(\omega)| = \frac{R/L}{\sqrt{(R/L)^2 + \omega_c^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c = \frac{R}{L}$$

11.6 Highpass filters (f)

- Example 11.01:

Example 11.01: Determine the transfer function and cutoff frequency in the following circuit when the load resistance is (1) R_L and (2) R .



Solution: (1) The load resistance is equal to R_L

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{\frac{j\omega L R_L}{R_L + j\omega L}}{R + \frac{j\omega L R_L}{R_L + j\omega L}} = \frac{\frac{R_L}{R + R_L} j\omega}{j\omega + \frac{R_L}{R + R_L} \frac{R}{L}} = \frac{j\omega K}{j\omega + K \frac{R}{L}} = \frac{j\omega}{\frac{j\omega}{K} + \frac{R}{L}}, \quad K = \frac{R_L}{R + R_L}$$

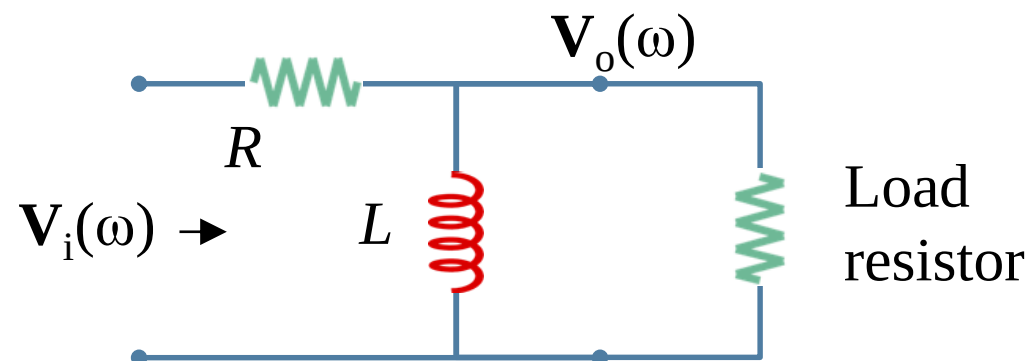
$$\text{Let } |\mathbf{H}(\omega)| = \frac{1}{\sqrt{2}}, \quad \omega_c = \frac{KR}{L}$$

How does the magnitude change after a load resistor is connected?

11.6 Highpass filters (g)

- Example 11.01:

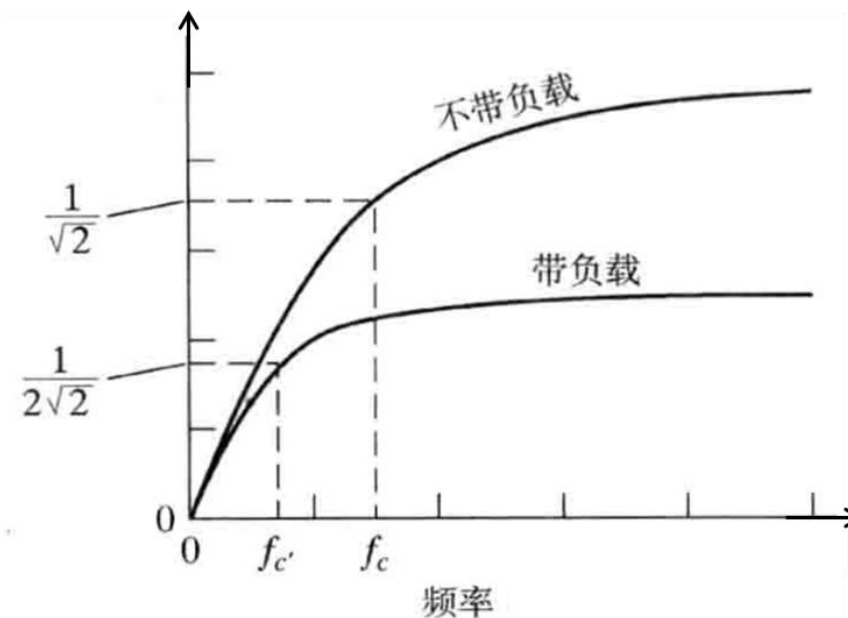
Example 11.01: Determine the transfer function and cutoff frequency in the following circuit when the load resistance is (1) R_L and (2) R .



Solution: (2) The load resistance is equal to R

$$K = \frac{R}{R + R} = 0.5, \mathbf{H}(\omega) = \frac{j\omega}{j\omega + \frac{2R}{L}} = \frac{j\omega 0.5}{\frac{j\omega}{0.5} + \frac{R}{L}}$$

$$\omega_c = \frac{0.5R}{L}$$



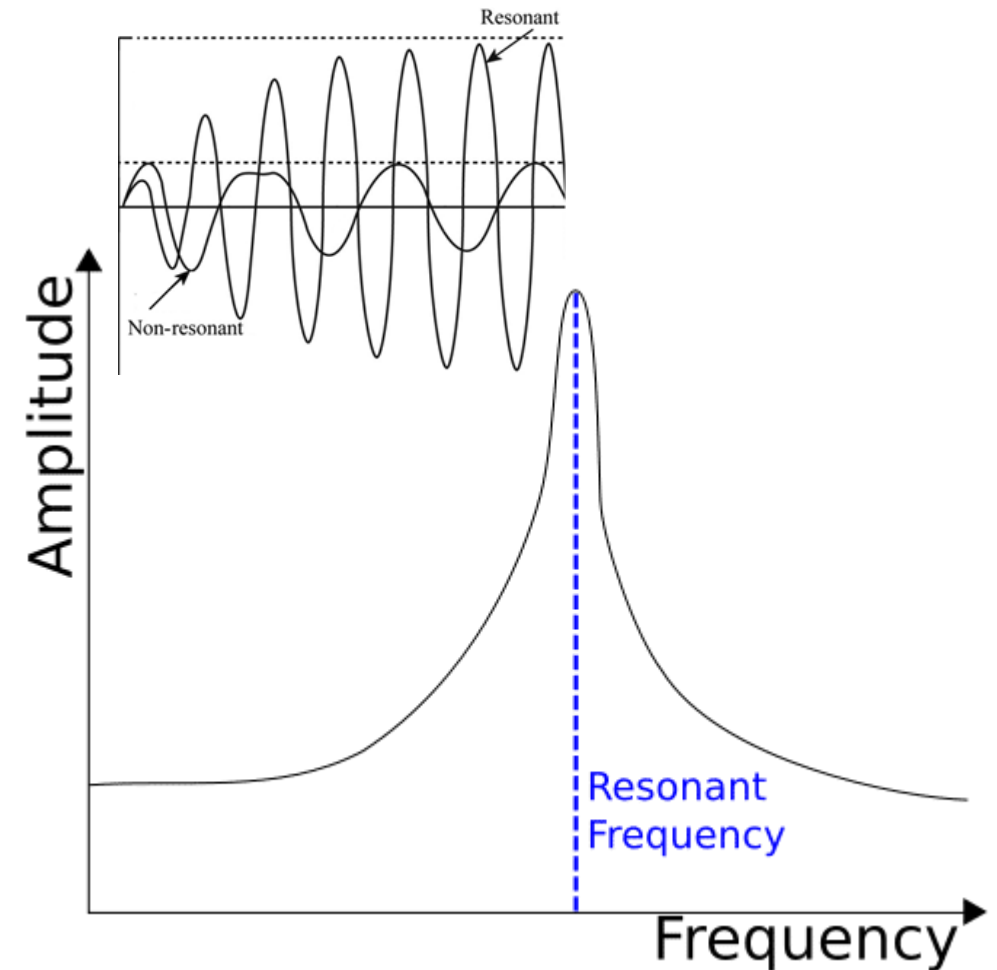
Lesson 11 Frequency response

11.1-11.4	Transfer function, The Decibel Scale, Bode Plots	Slides 09-20
11.5-11.6	Lowpass and highpass filters	Slides 22-37
11.7-11.8	Series and parallel resonances	Slides 39-54
11.9	Bandpass and bandstop filters	Slides 56-66
11.10	Active filters	Slides 68-75
11.11	High-order filters and scaling	Slides 77-81
11.12	Summary	Slides 83-85

11.7 Series resonance (a)

- Resonance (谐振):

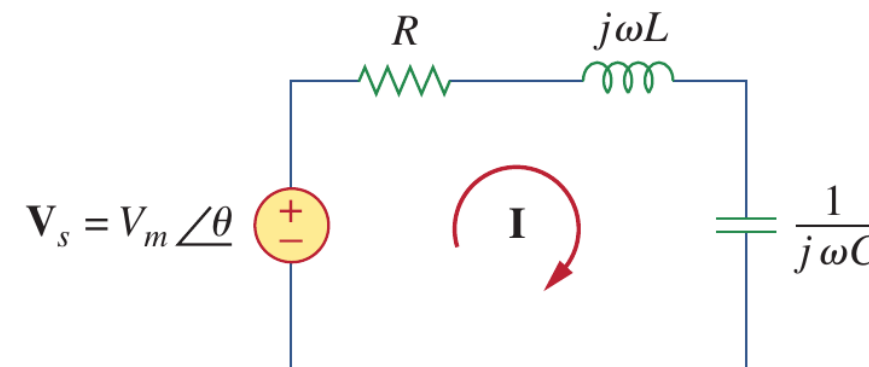
- a) The most prominent feature of the frequency response of a circuit maybe the sharp peak (or resonant peak) exhibited in its amplitude characteristic.
- b) Resonance is the cause of oscillations of stored energy from one form to another.
- c) It is the phenomenon that allows frequency discrimination in communications networks.



11.7 Series resonance (b)

- Resonance (谐振) and resonant frequency (谐振频率):

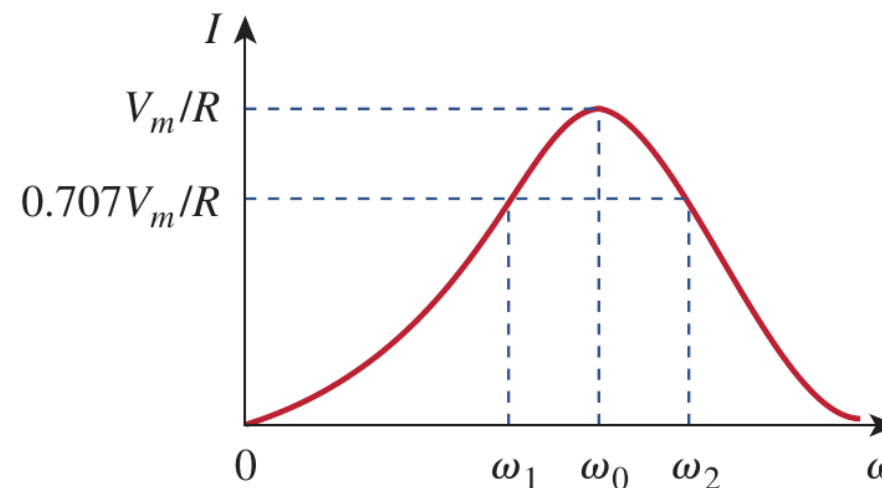
- a) Resonance is a condition in an RLC circuit in which the capacitive and inductive reactances are equal in magnitude, resulting in a purely resistive impedance.
- b) The corresponding frequency is called resonant frequency ω_0 .



$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

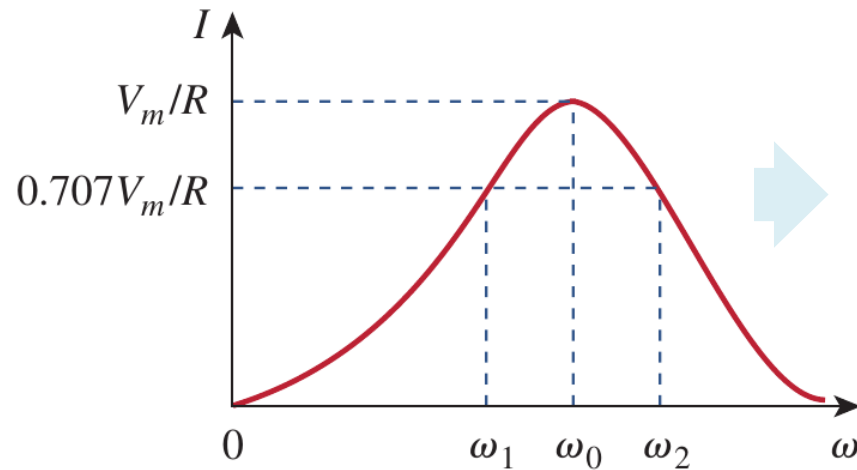
$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

$$\text{Let } \text{Im}(\mathbf{Z}) = \omega L - \frac{1}{\omega C} = 0 \quad \omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$



11.7 Series resonance (c)

- Cutoff frequency (ω_1 and ω_2) and bandwidth (B):



$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

*Only for series RLC circuits

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$B = \omega_2 - \omega_1$$

- The **cutoff frequencies** (ω_1 and ω_2) are the half-power frequency as explained in section 11.5.
- The **bandwidth** (B) is defined as the difference between the two half-power frequencies. This definition of bandwidth is just one of several that are commonly used. Strictly speaking, is a half-power bandwidth, because it is the width of the frequency band between the half-power frequencies.

11.7 Series resonance (d)

- Quality factor (Q):

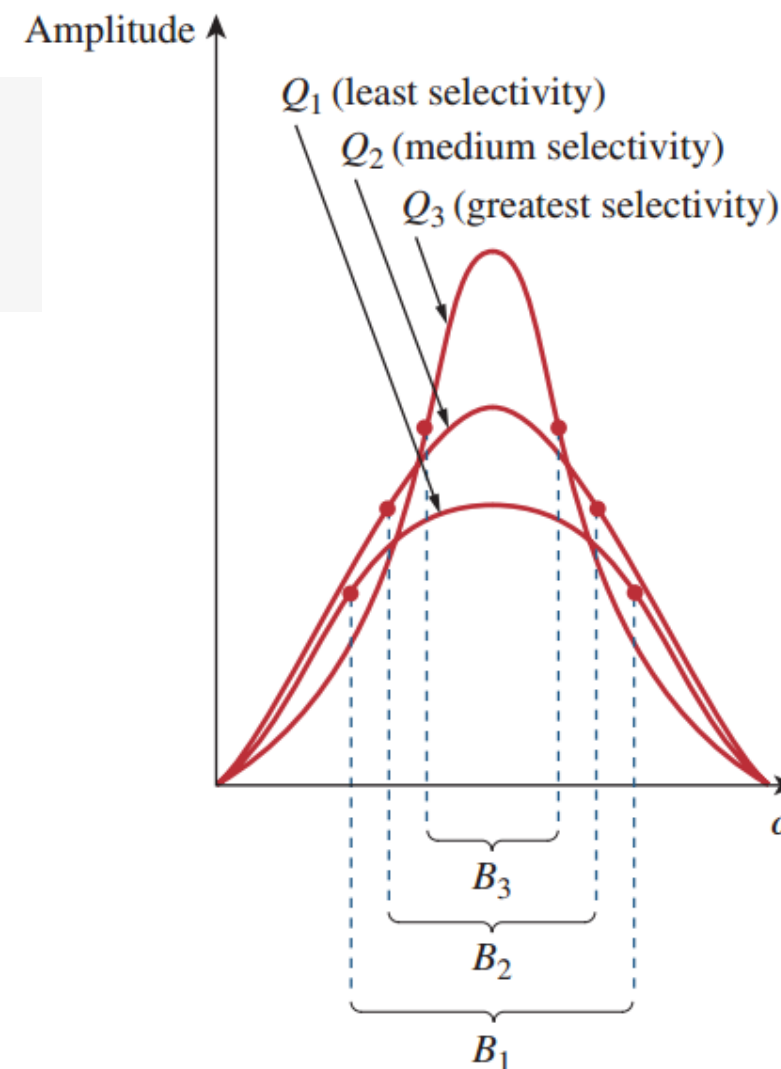
- a) The **quality factor (Q)** is defined as the ratio between the peak energy stored in the circuit and the energy dissipated by the resistor within one resonant period.

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

$$Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0 L}{R}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

*Only for series RLC circuits



11.7 Series resonance (e)

- Quality factor (Q) and bandwidth (B):

We know that:

$$B = \omega_2 - \omega_1$$

For series RLC circuits

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$$

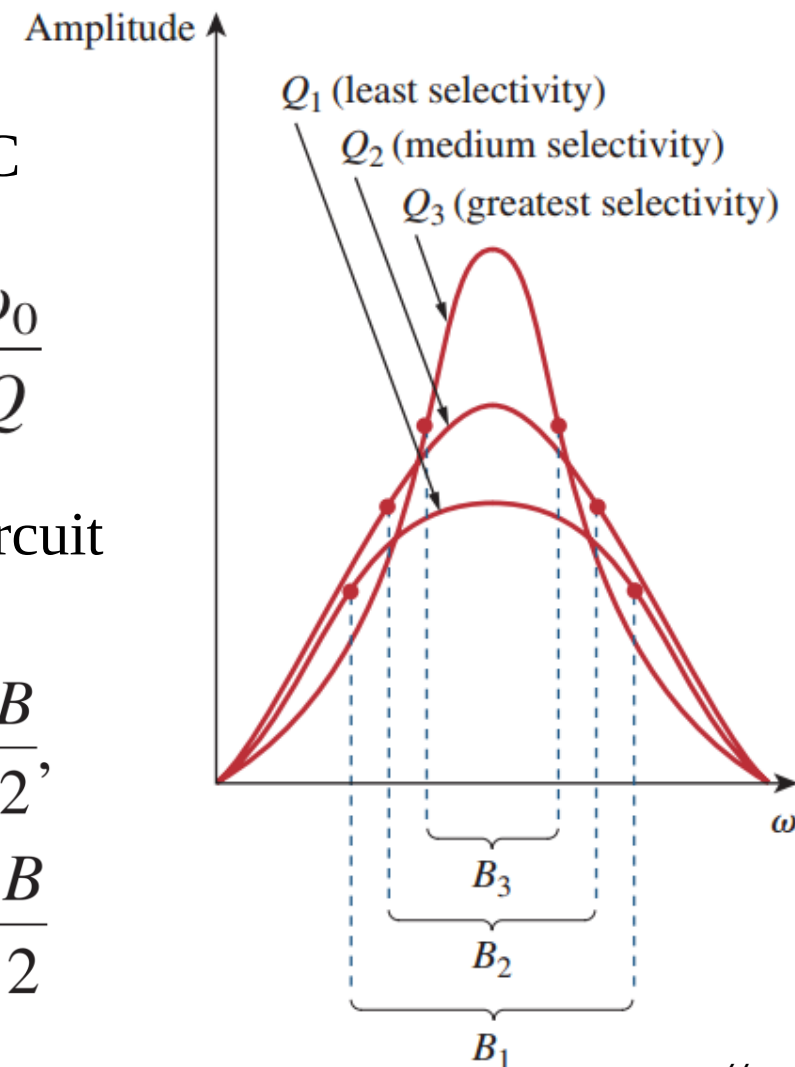
For series RLC circuits:

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

For high- Q circuit ($Q > 10$):

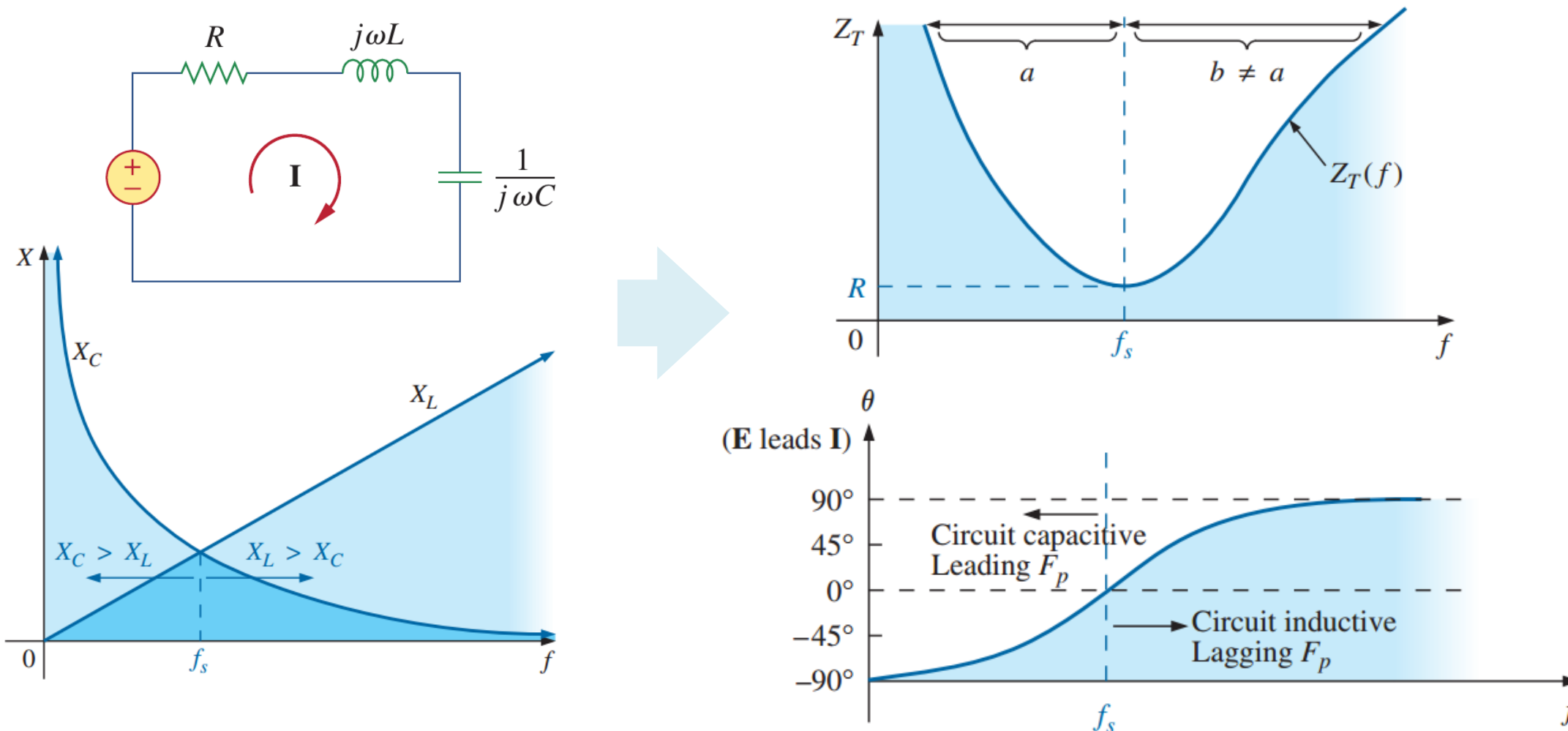
$$\omega_1 \approx \omega_0 - \frac{B}{2},$$

$$\omega_2 \approx \omega_0 + \frac{B}{2}$$



11.7 Series resonance (f)

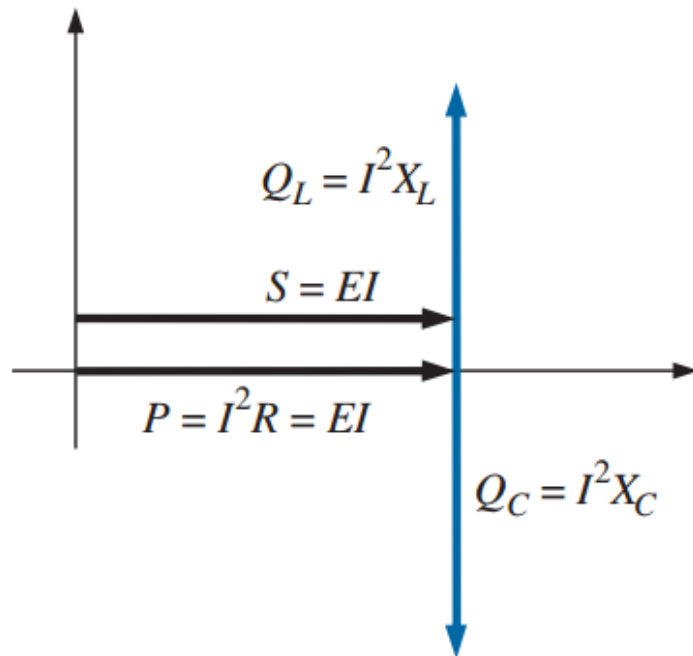
- Again it is all about the impedance:



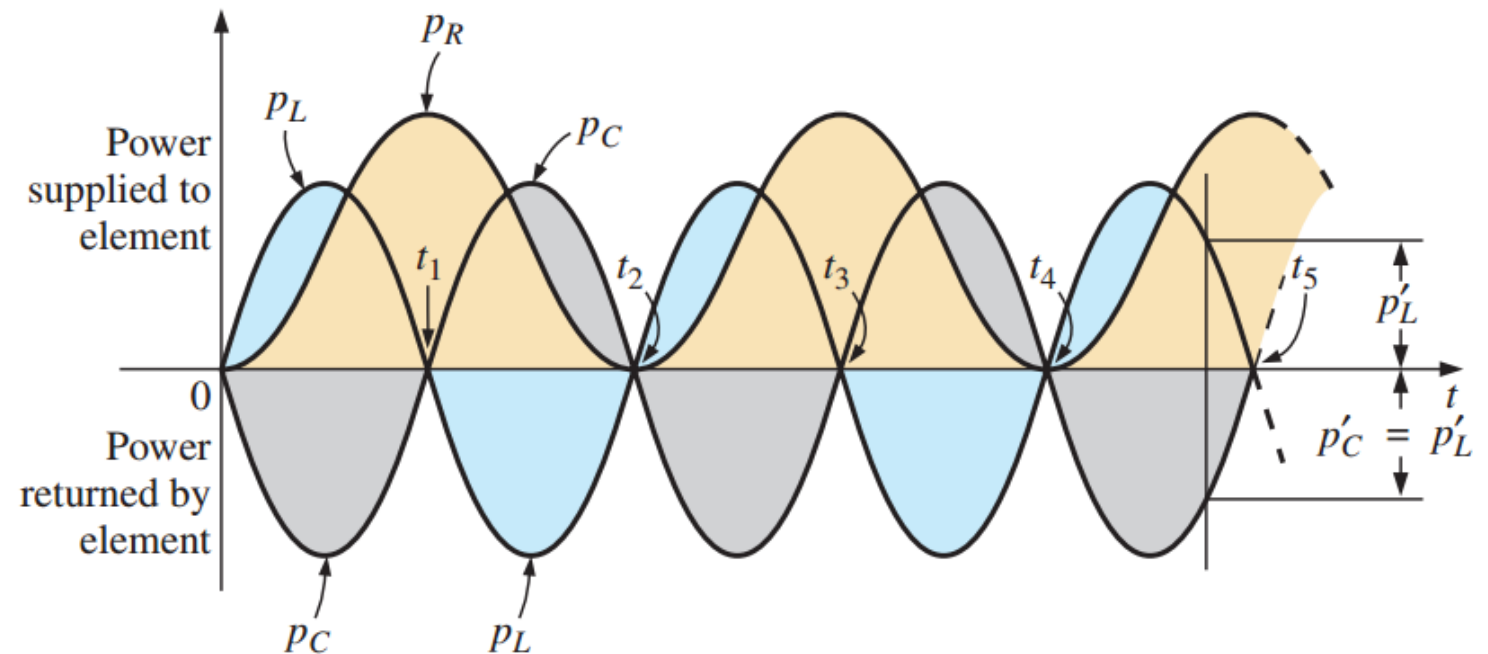
11.7 Series resonance (g)

- The power in series resonance:

Power triangle:



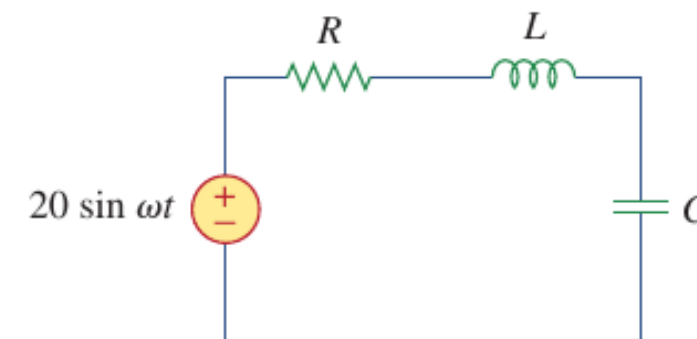
Instantaneous power:



11.7 Series resonance (h)

- **Example 11.02:**

Example 11.02: In the following circuit, $R = 2 \Omega$, $L = 1 \text{ mH}$, and $C = 0.4 \text{ mF}$. (a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0 , ω_1 , and ω_2 .



Solution I: (a)
$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}} = 50 \text{ krad/s}$$

(b)
$$\begin{aligned} \omega_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \\ &= -\frac{2}{2 \times 10^{-3}} + \sqrt{(10^3)^2 + (50 \times 10^3)^2} \\ &= -1 + \sqrt{1 + 2500} \text{ krad/s} = 49 \text{ krad/s} \end{aligned}$$

$$\omega_2 = 1 + \sqrt{1 + 2500} \text{ krad/s} = 51 \text{ krad/s}$$

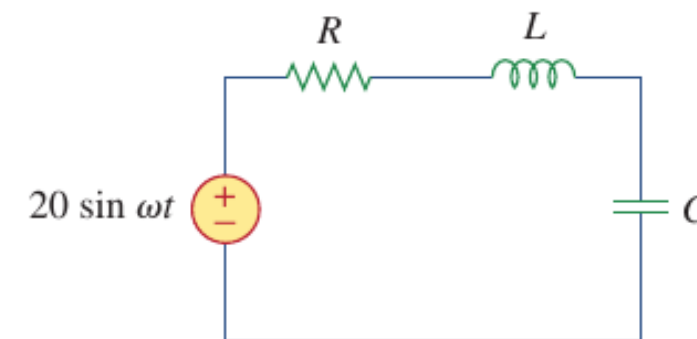
$$B = \omega_2 - \omega_1 = 2 \text{ krad/s} \quad B = \frac{R}{L} = \frac{2}{10^{-3}} = 2 \text{ krad/s}$$

$$Q = \frac{\omega_0}{B} = \frac{50}{2} = 25$$

11.7 Series resonance (i)

- Example 11.02:

Example 11.02: In the following circuit, $R = 2\ \Omega$, $L = 1\ \text{mH}$, and $C = 0.4\ \text{mF}$. (a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0 , ω_1 , and ω_2 .



Solution II: (a)
$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}} = 50\ \text{krad/s}$$

(b)
$$Q = \frac{\omega_0 L}{R} = \frac{50 \times 10^3 \times 10^{-3}}{2} = 25 \quad B = \frac{\omega_0}{Q} = \frac{50 \times 10^3}{25} = 2\ \text{krad/s}$$

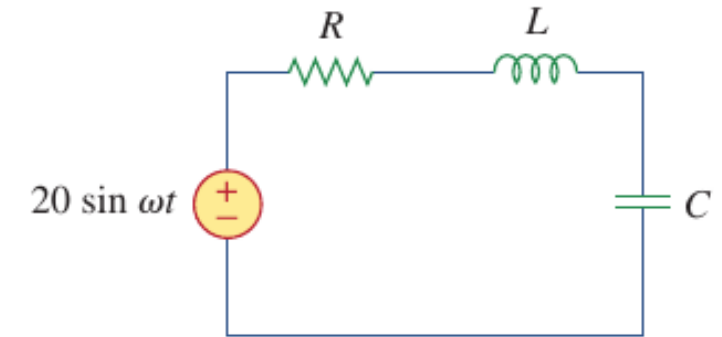
$$\omega_1 = \omega_0 - \frac{B}{2} = 50 - 1 = 49\ \text{krad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 50 + 1 = 51\ \text{krad/s}$$

11.7 Series resonance (j)

- Example 11.02:

Example 11.02: In the following circuit, $R = 2\ \Omega$, $L = 1\ \text{mH}$, and $C = 0.4\ \text{mF}$. (a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0 , ω_1 , and ω_2 .



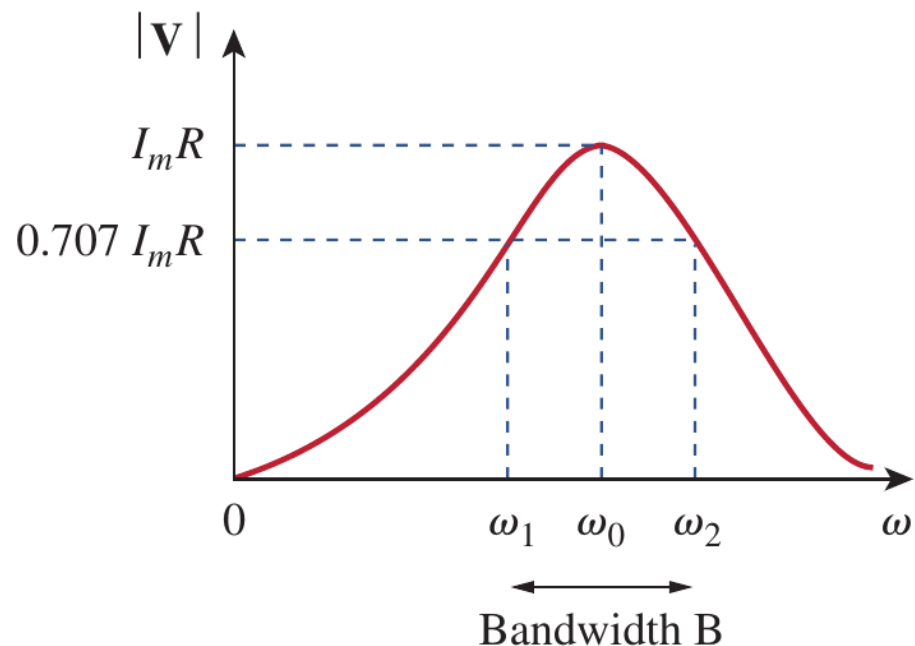
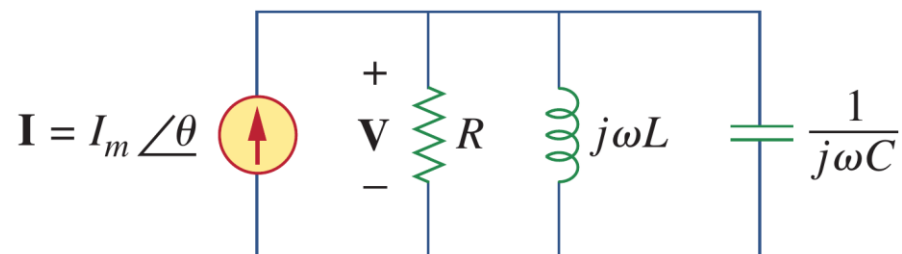
Solution:

$$(c) \quad \text{at } \omega_0 \quad I = \frac{V_m}{R} = \frac{20}{2} = 10\ \text{A}$$

$$\text{at } \omega_1 \text{ and } \omega_2 \quad I = \frac{V_m}{\sqrt{2}R} = \frac{10}{\sqrt{2}} = 7.071\ \text{A}$$

11.8 Parallel resonance (a)

- Parallel resonance (并联谐振):



$$\mathbf{Y} = H(\omega) = \frac{\mathbf{I}}{\mathbf{V}} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

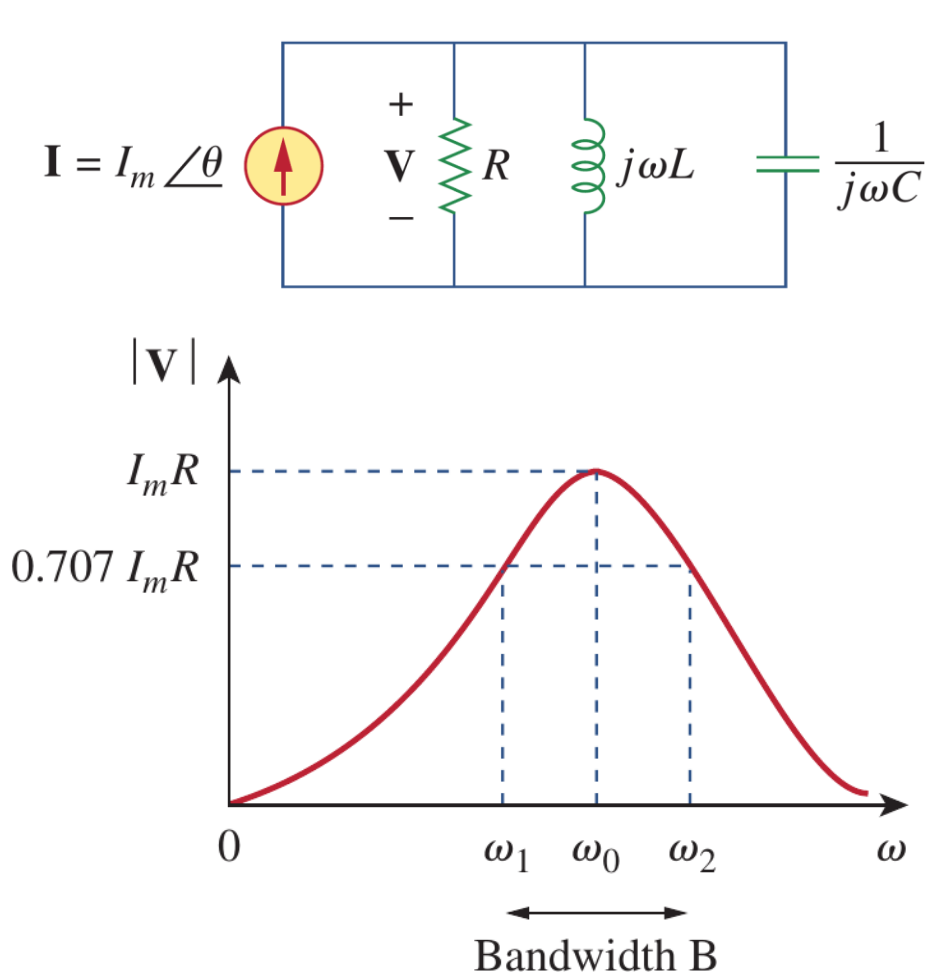
$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$B = \omega_2 - \omega_1 = \frac{1}{RC}$$

11.8 Parallel resonance (b)

- Parallel resonance (并联谐振):



$$Q = \frac{\omega_0}{B} = \omega_0 RC = \frac{R}{\omega_0 L}$$

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} = \omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} - \frac{\omega_0}{2Q}$$

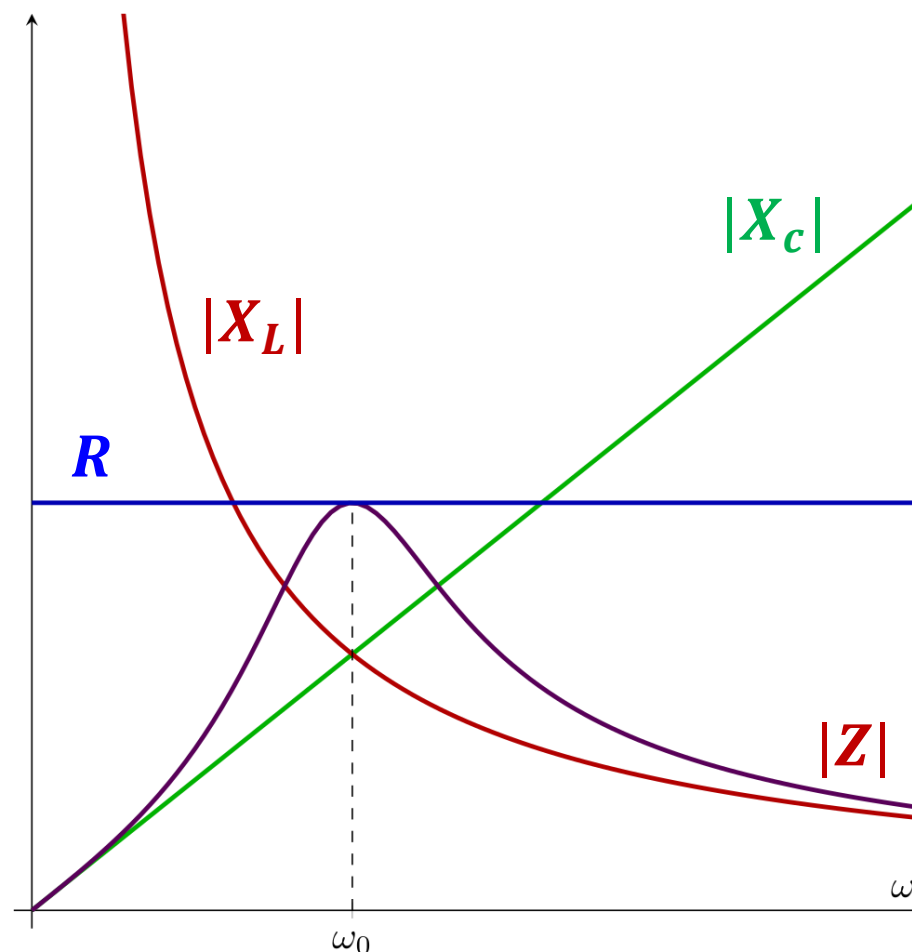
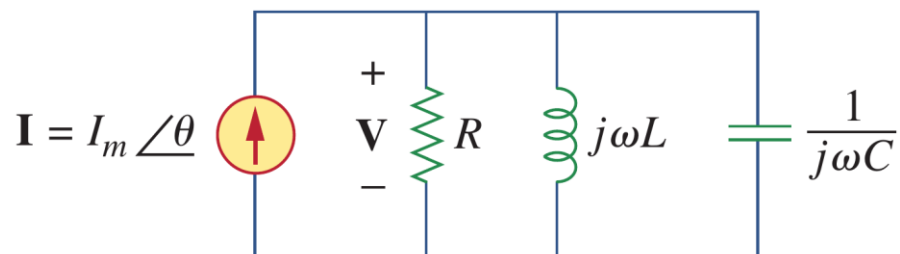
$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} = \omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} + \frac{\omega_0}{2Q}$$

For high-Q circuit ($Q > 10$):

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \quad \omega_2 \simeq \omega_0 + \frac{B}{2}$$

11.8 Parallel resonance (c)

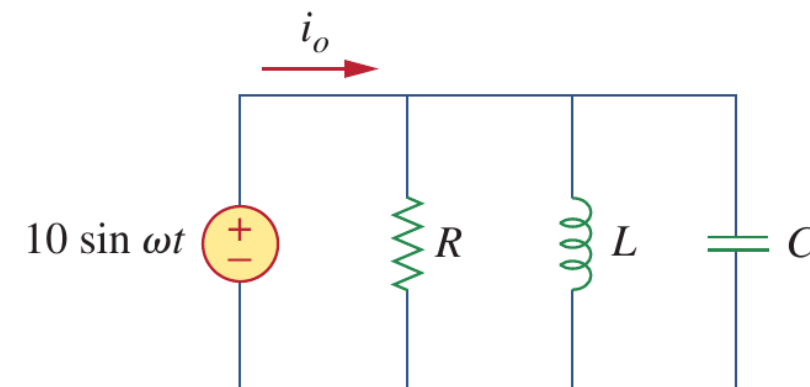
- Parallel resonance (并联谐振):



11.8 Parallel resonance (d)

- Example 11.03:

Example 11.03: $R = 8 \text{ k}\Omega$, $L = 0.2 \text{ mH}$, and $C = 8 \text{ mF}$. (a) Calculate ω_0 , Q , and B . (b) Find ω_1 and ω_2 . (c) Determine the power dissipated at ω_0 , ω_1 , and ω_2 .



Solution: (a) $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.2 \times 10^{-3} \times 8 \times 10^{-6}}} = \frac{10^5}{4} = 25 \text{ krad/s}$

$$Q = \frac{R}{\omega_0 L} = \frac{8 \times 10^3}{25 \times 10^3 \times 0.2 \times 10^{-3}} = 1,600$$

$$B = \frac{\omega_0}{Q} = 15.625 \text{ rad/s}$$

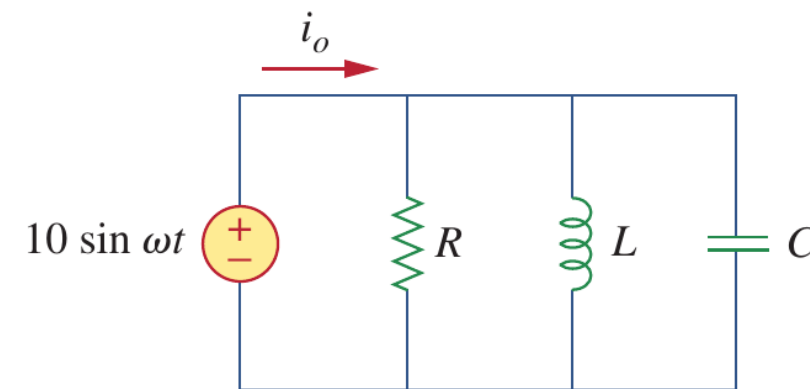
(b) Due to the high quality factor:

$$\omega_1 = \omega_0 - \frac{B}{2} = 25,000 - 7.812 = 24,992 \text{ rad/s} \quad \omega_2 = \omega_0 + \frac{B}{2} = 25,000 + 7.812 = 25,008 \text{ rad/s}$$

11.8 Parallel resonance (e)

- Example 11.03:

Example 11.03: $R = 8 \text{ k}\Omega$, $L = 0.2 \text{ mH}$, and $C = 8 \text{ mF}$. (a) Calculate ω_0 , Q , and B . (b) Find ω_1 and ω_2 . (c) Determine the power dissipated at ω_0 , ω_1 , and ω_2 .



Solution: (c) At $\omega = \omega_0$, $\mathbf{Y} = 1/R$ or $\mathbf{Z} = R = 8 \text{ k}\Omega$. Then

$$\mathbf{I}_o = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{10 \angle -90^\circ}{8,000} = 1.25 \angle -90^\circ \text{ mA}$$

$$P = \frac{V_m^2}{2R} = \frac{100}{2 \times 8 \times 10^3} = 6.25 \text{ mW}$$

$$\text{At } \omega = \omega_1, \omega_2, \quad P = \frac{V_m^2}{4R} = 3.125 \text{ mW}$$

11.8 Parallel resonance (f)

- Comparison of series and parallel resonances:

Characteristic	Series circuit	Parallel circuit
Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$

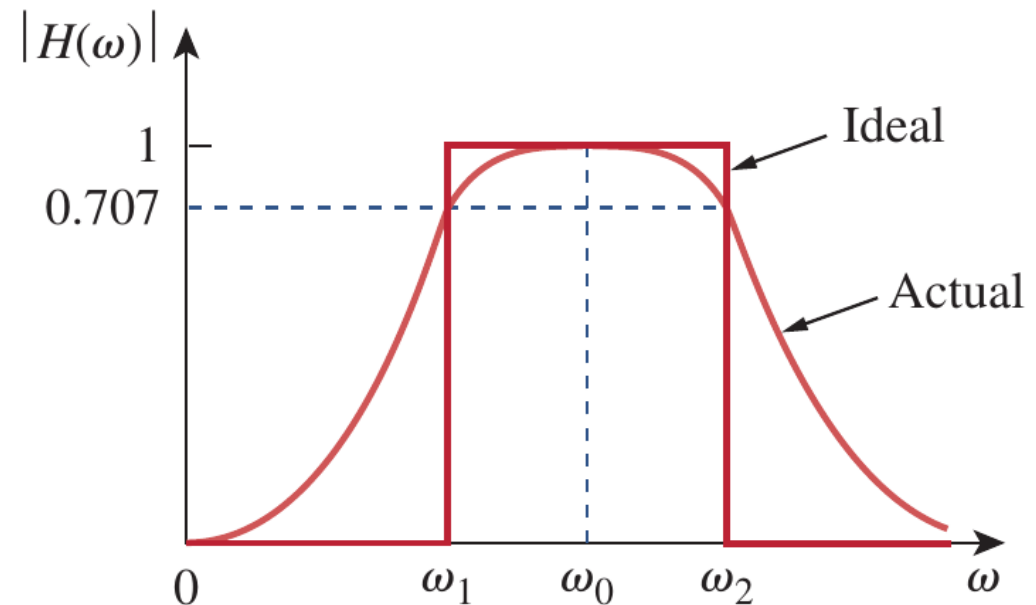
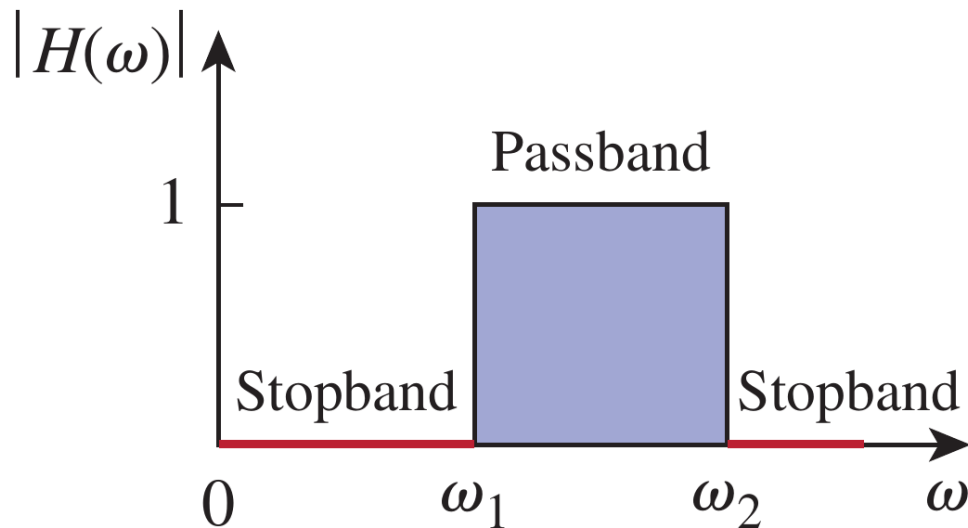
Lesson 11 Frequency response

11.1-11.4	Transfer function, The Decibel Scale, Bode Plots	Slides 09-20
11.5-11.6	Lowpass and highpass filters	Slides 22-37
11.7-11.8	Series and parallel resonances	Slides 39-54
11.9	Bandpass and bandstop filters	Slides 56-66
11.10	Active filters	Slides 68-75
11.11	High-order filters and scaling	Slides 77-81
11.12	Summary	Slides 83-85

11.9 Bandpass and bandstop filters (a)

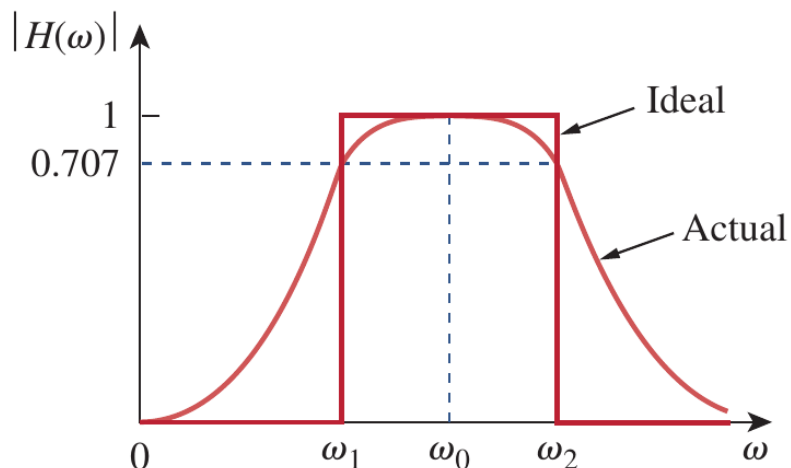
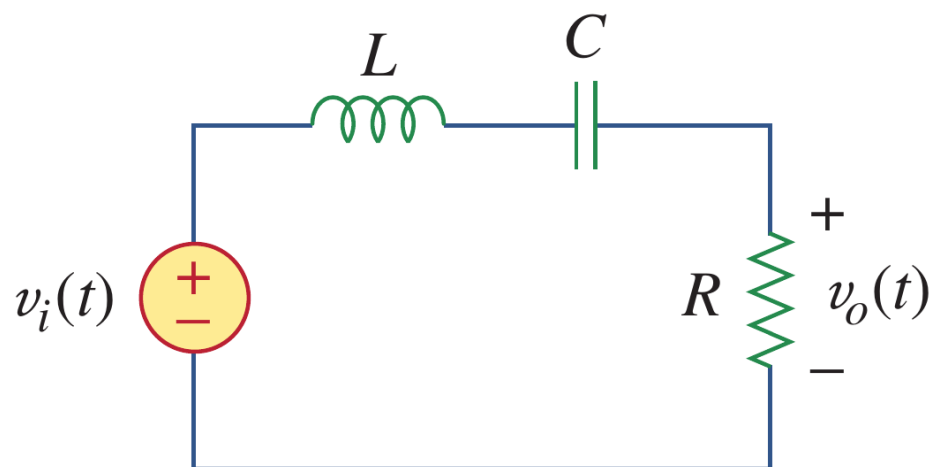
- Bandpass filters (带通滤波器):

a) A **bandpass filter** is designed to pass all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.



11.9 Bandpass and bandstop filters (b)

- RLC Bandpass filters (带通滤波器):



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The filter will pass frequencies from ω_1 to ω_2 .

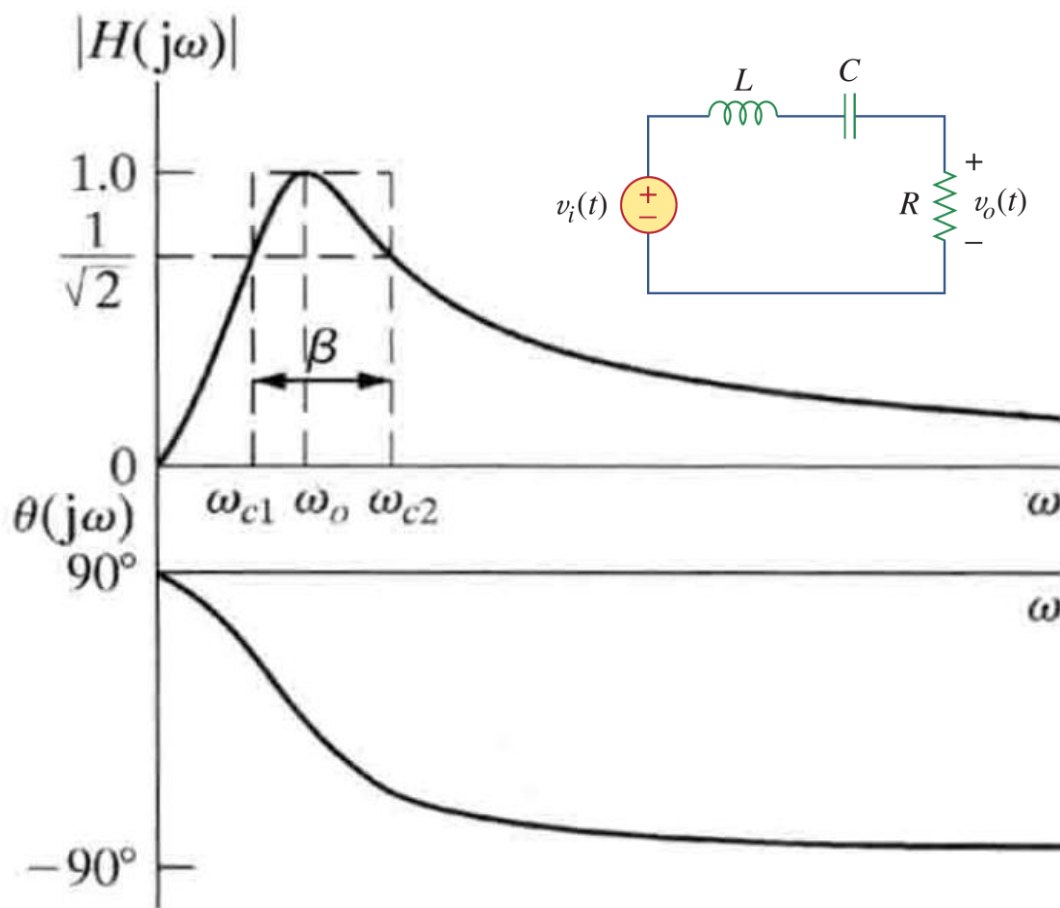
ω_1 : lower cut-off frequency 下限截止频率

ω_2 : upper cut-off frequency 上限截止频率

It can also be made by feeding the output of a lowpass to a highpass filter.

11.9 Bandpass and bandstop filters (c)

- RLC Bandpass filters (带通滤波器):



$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s} \quad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$B = \omega_2 - \omega_1 \quad B = \frac{R}{L} = \frac{\omega_0}{Q}$$

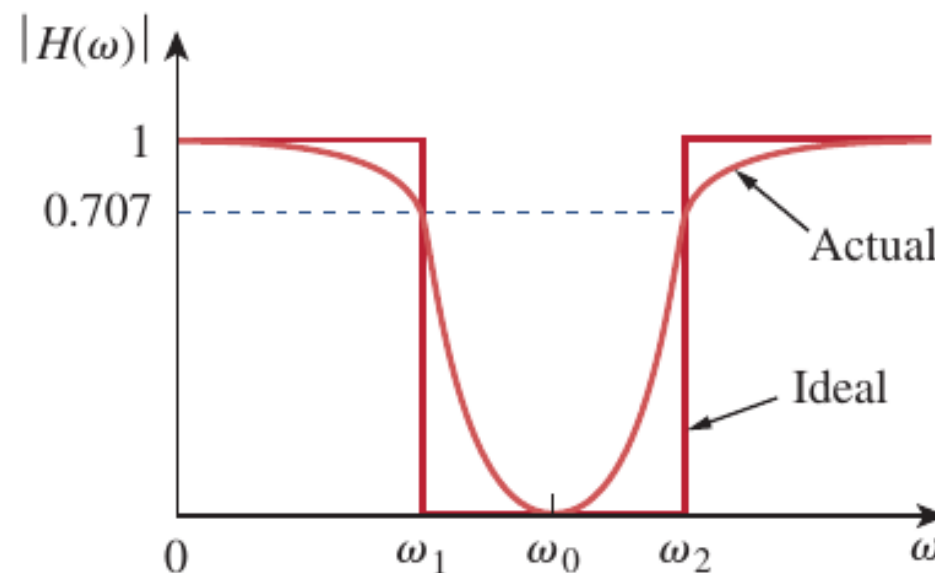
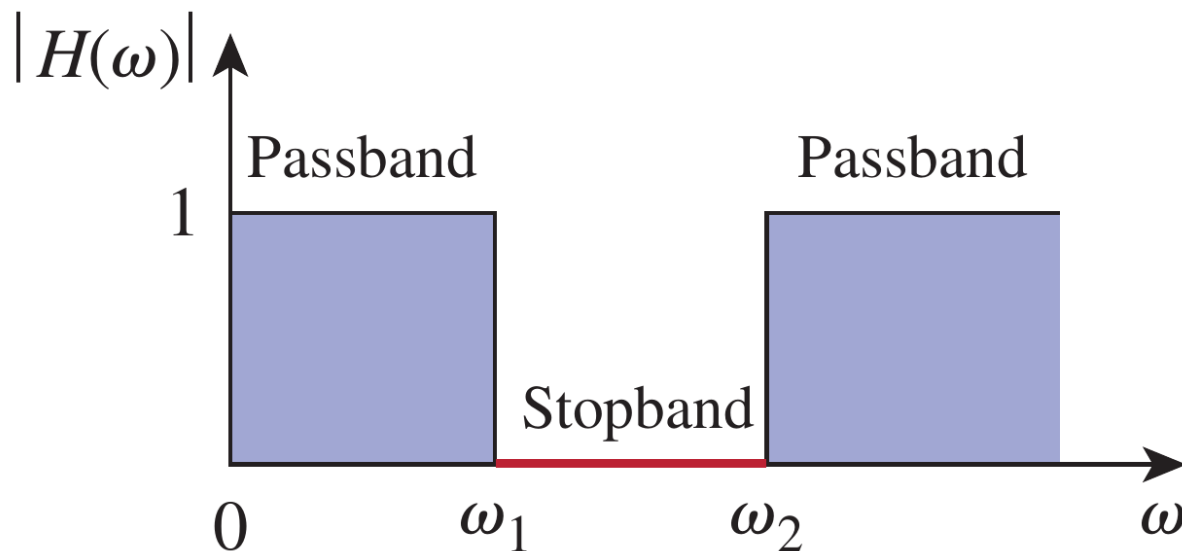
For high-Q circuit ($Q > 10$):

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \quad \omega_2 \simeq \omega_0 + \frac{B}{2}$$

11.9 Bandpass and bandstop filters (d)

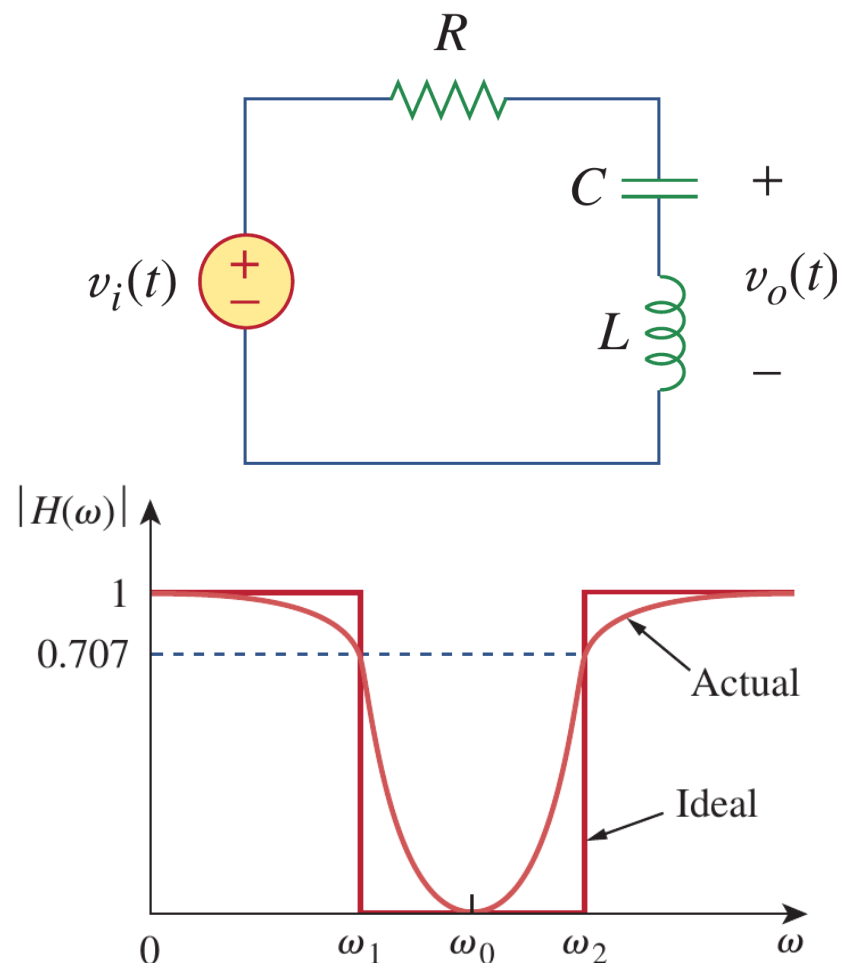
- Bandstop filters (帶阻濾波器):

- a) A **bandstop filter** is designed to stop, reject or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.



11.9 Bandpass and bandstop filters (e)

- RLC Bandstop filters (带阻滤波器):



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

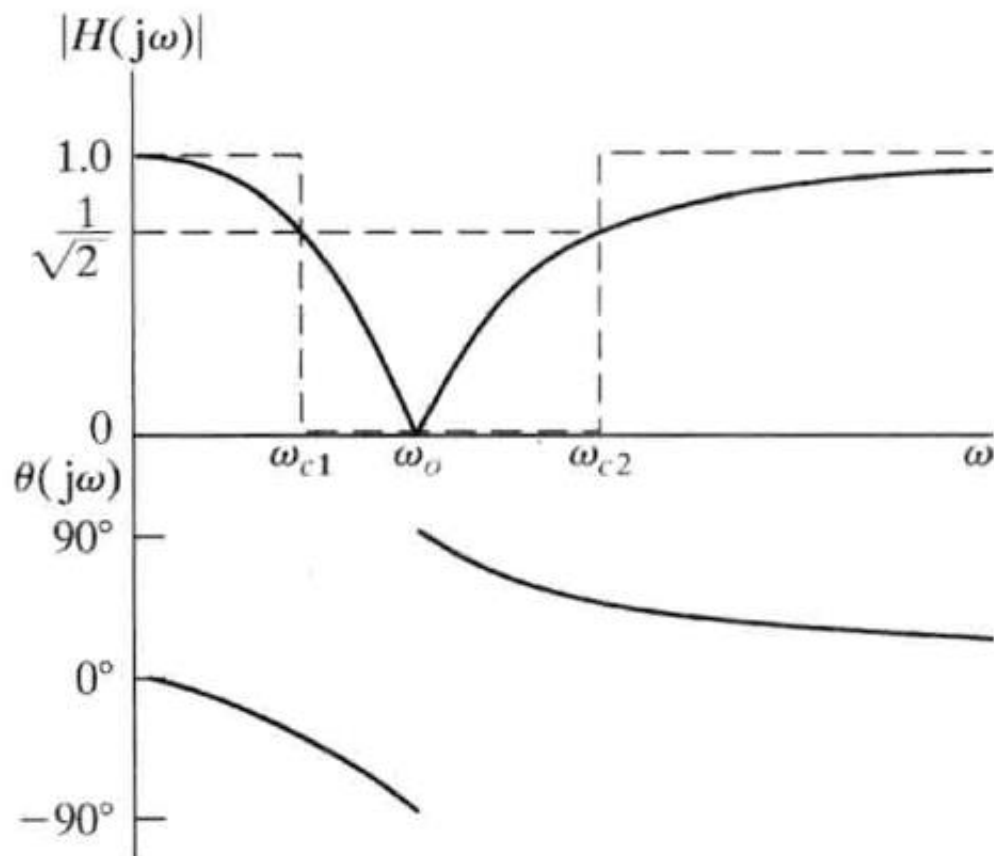
- a) ω_0 is also called frequency of rejection 抑制频率
- b) $B = \omega_2 - \omega_1$ is called bandwidth of rejection 抑制带宽

$$|H(\omega)| = \left| \frac{1}{LC} - \omega^2 \right| / \sqrt{\left(\frac{1}{LC} - \omega^2 \right)^2 + \left(\frac{\omega R}{L} \right)^2}$$

$$\theta = -\tan^{-1} \left(\frac{\omega R}{L} / \left(\frac{1}{LC} - \omega^2 \right) \right)$$

11.9 Bandpass and bandstop filters (f)

- Bandstop filters (帶阻濾波器):



$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s} \quad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$B = \omega_2 - \omega_1 \quad B = \frac{R}{L} = \frac{\omega_0}{Q}$$

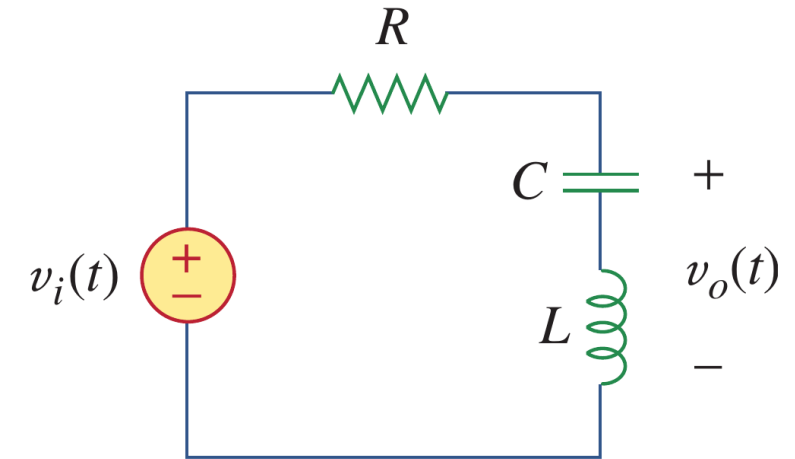
For high-Q circuit ($Q > 10$):

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \quad \omega_2 \simeq \omega_0 + \frac{B}{2}$$

11.9 Bandpass and bandstop filters (g)

- Example 11.04:

Example 11.04: If the bandstop filter in the following circuit is to reject a 200-Hz sinusoid while passing other frequencies, calculate the values of L and C . Take $R = 150\ \Omega$ and the bandwidth as 100 Hz.



Solution:

The bandwidth: $B = 2\pi(100) = 200\pi\text{ rad/s}$

The inductance: $B = \frac{R}{L} \Rightarrow L = \frac{R}{B} = \frac{150}{200\pi} = 0.2387\text{ H}$

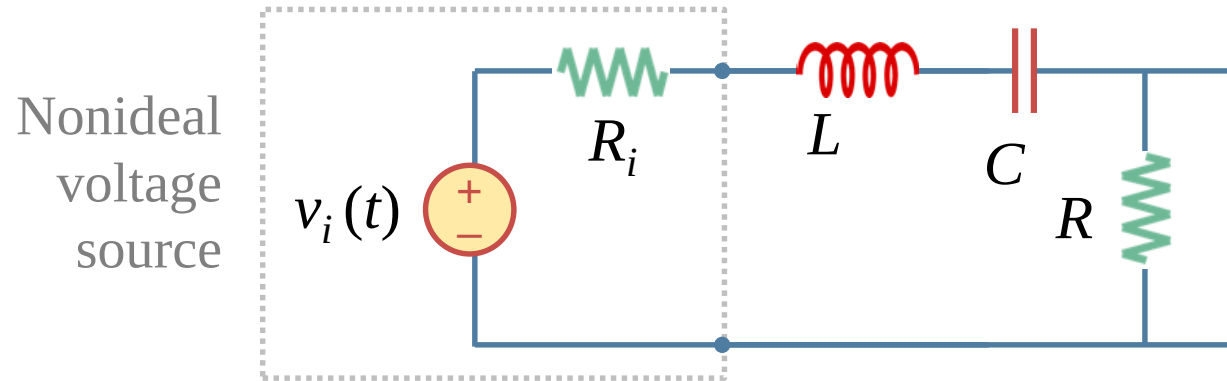
As the filter rejects a 200 Hz sinusoid while pass others,

$$\omega_0 = 2\pi f_0 = 2\pi(200) = 400\pi$$

$$\text{Since } \omega_0 = 1/\sqrt{LC}, \quad C = \frac{1}{\omega_0^2 L} = \frac{1}{(400\pi)^2(0.2387)} = 2.653\ \mu\text{F}$$

11.9 Bandpass and bandstop filters (h)

- Example 11.05:



Example 11.05: For each of the bandpass filters we have assumed ideal voltage sources without internal resistance. Even though this assumption is often valid, sometimes it is not, as in the case where the filter design can be achieved only with values of R , L , C whose equivalent impedance has a magnitude close to the actual impedance of the voltage sources. Examine the effect of assuming a nonzero source resistance, R_i , on the characteristics of a series RLC bandpass filter.

11.9 Bandpass and bandstop filters (i)

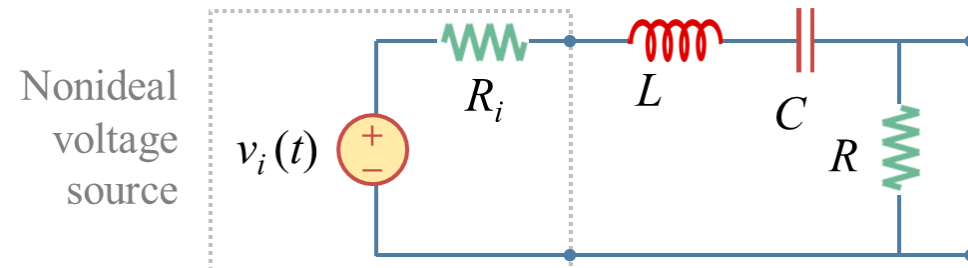
- Example 11.05:

Solution: $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\mathbf{H}(\omega) = \frac{R}{R_i + R + j\omega L + \frac{1}{j\omega C}} = \frac{\frac{R}{L} j\omega}{(j\omega)^2 + j\omega \left(\frac{R_i + R}{L} \right) + \frac{1}{LC}}$$

$$|\mathbf{H}(\omega)| = \frac{\frac{R}{L} \omega}{\sqrt{\left(\frac{1}{LC} - \omega^2 \right)^2 + \left(\omega \frac{R + R_i}{L} \right)^2}}, \quad H_{max} = |\mathbf{H}(\omega_0)| = \frac{R}{R + R_i}$$

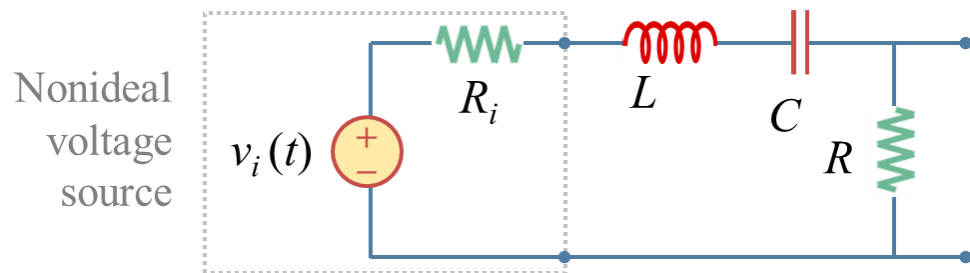
$$\omega_1 = -\frac{R + R_i}{2L} + \sqrt{\left(\frac{R + R_i}{2L} \right)^2 + \frac{1}{LC}}, \quad \text{and} \quad \omega_2 = +\frac{R + R_i}{2L} + \sqrt{\left(\frac{R + R_i}{2L} \right)^2 + \frac{1}{LC}},$$



11.9 Bandpass and bandstop filters (j)

- Example 11.05:

Solution:

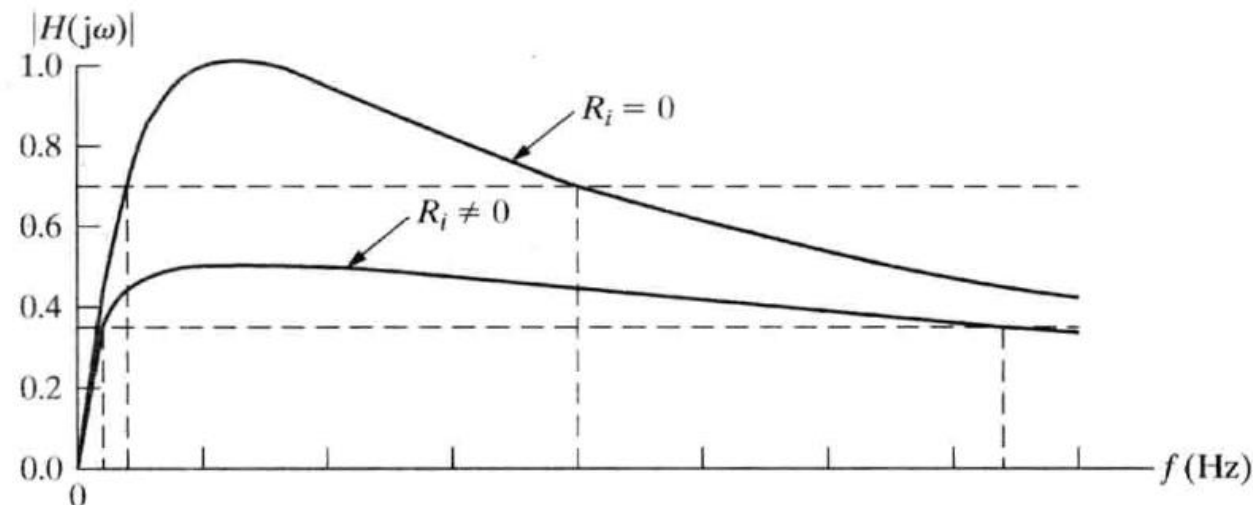


With R_i

$$B = \frac{R + R_i}{L} \quad Q = \frac{\sqrt{L/C}}{R + R_i}$$

Without R_i

$$B = \frac{R}{L} \quad Q = \frac{\sqrt{L/C}}{R}$$



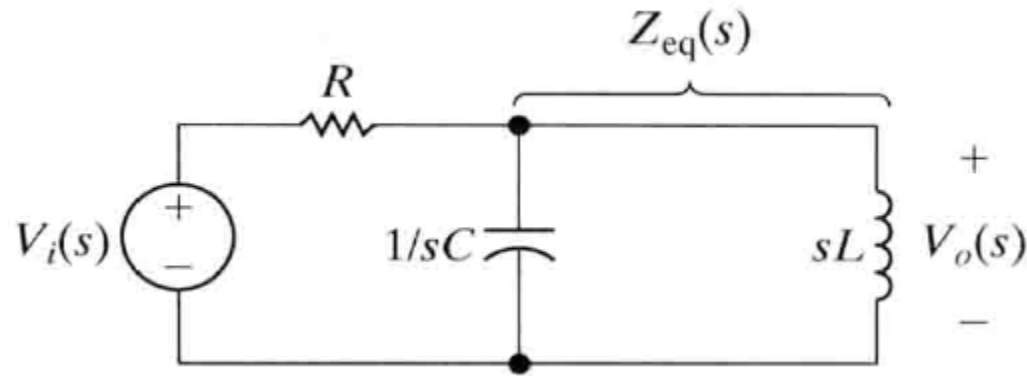
With nonideal voltage sources (with R_i):

The same ω_0

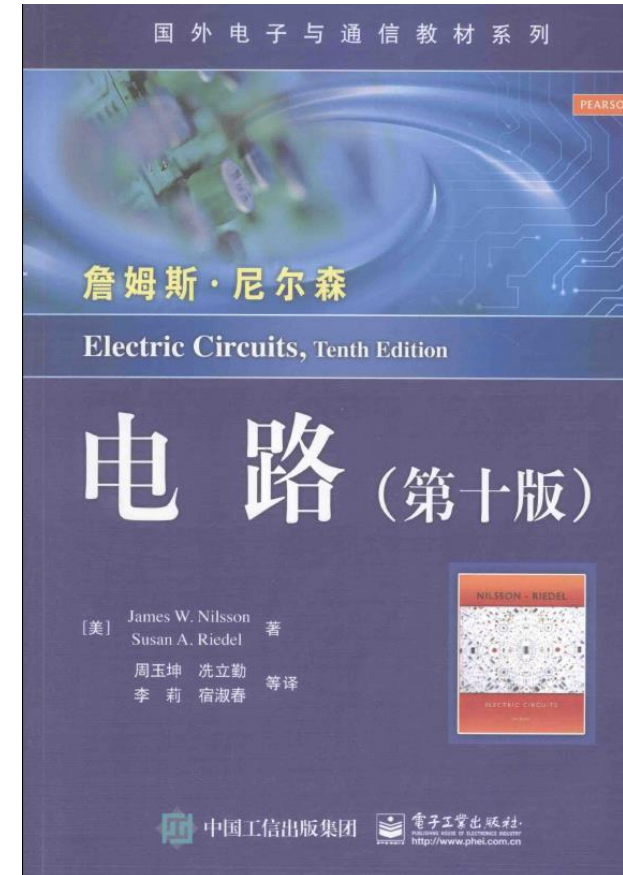
Smaller Q and larger B

11.9 Bandpass and bandstop filters (k)

- Parallel RCL circuits can also be bandpass filters:



Check Example 14.6 on pages
440 – 442 in Nilsson's book.



Lesson 11 Frequency response

11.1-11.4	Transfer function, The Decibel Scale, Bode Plots	Slides 09-20
11.5-11.6	Lowpass and highpass filters	Slides 22-37
11.7-11.8	Series and parallel resonances	Slides 39-54
11.9	Bandpass and bandstop filters	Slides 56-66
11.10	Active filters	Slides 68-75
11.11	High-order filters and scaling	Slides 77-81
11.12	Summary	Slides 83-85

11.10 Active filters (a)

- Advantages and disadvantages of passive filters:

Advantages:

- a) Simple, stable and reliable.
- b) They operate well in high frequencies.

Disadvantages:

- a) Passive filters cannot generate gain greater than 1, as they are passive elements and cannot add energy to the network.
- b) They may require bulky and expensive inductors.
- c) They perform poorly at frequencies below the audio frequency range (0.3 to 3 kHz).

11.10 Active filters (b)

- Advantages and disadvantages of active filters:

Advantages:

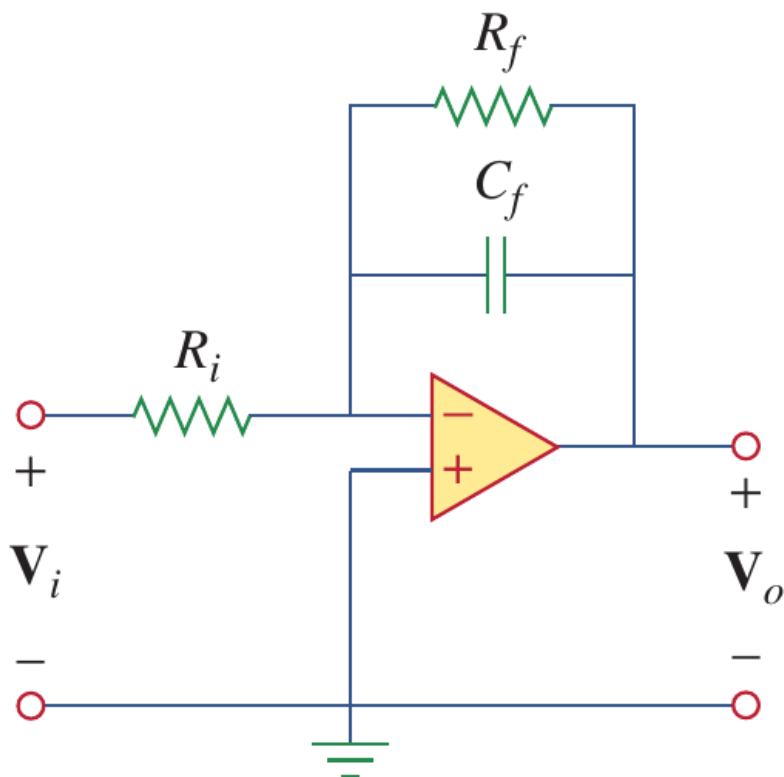
- a) Active filters can provide energy to the network and therefore generate gain greater than 1.
- b) They are often smaller and less expensive, because they do not require inductors.
- c) Active filters can be combined with buffer amplifiers (voltage followers) to isolate each stage of the filter from source and load impedance effects. This isolation allows designing the stages independently and then cascading them to realize the desired transfer function.

Disadvantages:

- a) Less reliable and stable.
- b) The practical limit for most active filters is 100 kHz, also they operate well below that frequency.

11.10 Active filters (c)

- Active lowpass filters:



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i}$$

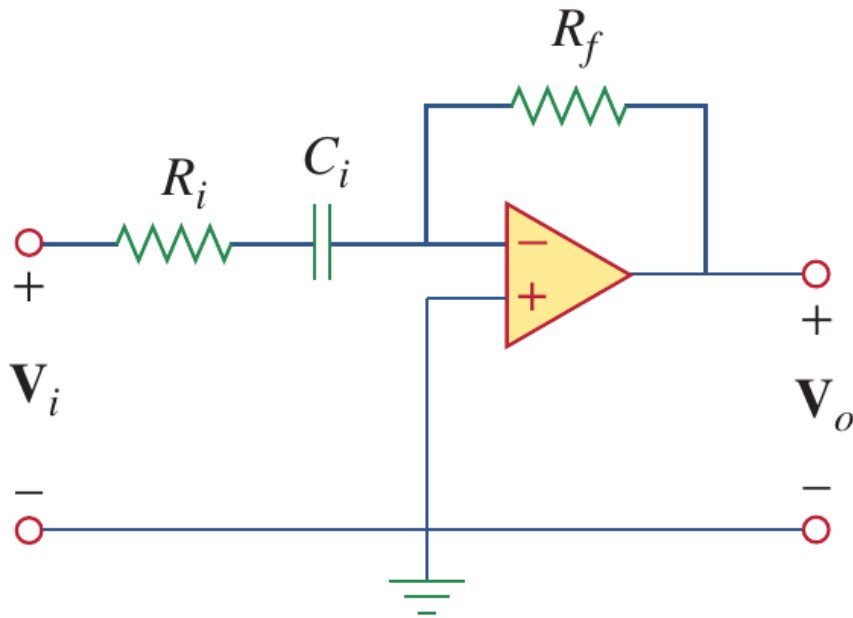
$$\mathbf{Z}_f = R_f \parallel \frac{1}{j\omega C_f} = \frac{R_f / j\omega C_f}{R_f + 1/j\omega C_f} = \frac{R_f}{1 + j\omega C_f R_f}$$

$$\omega \rightarrow 0, |\mathbf{H}(\omega)| \rightarrow -\frac{R_f}{R_i} \quad \omega \rightarrow \infty, |\mathbf{H}(\omega)| \rightarrow 0$$

$$\omega_c = \frac{1}{R_f C_f}$$

11.10 Active filters (d)

- Active highpass filters:



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i}$$

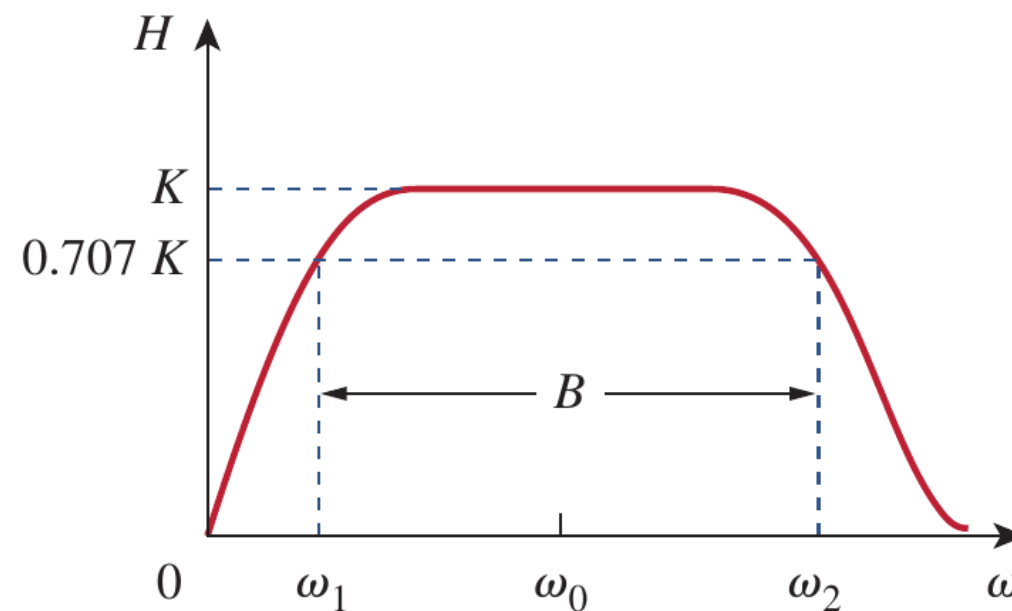
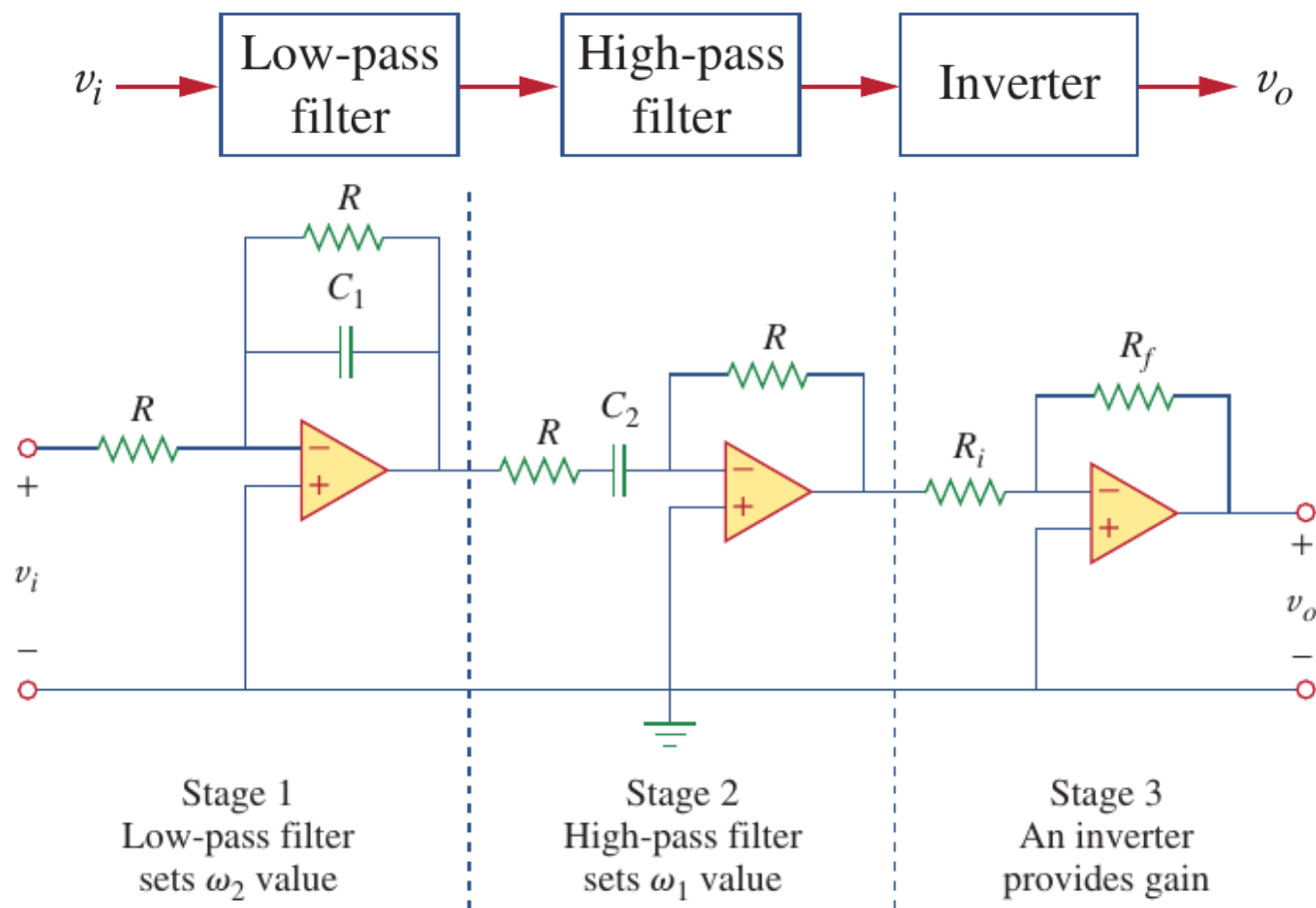
$$\mathbf{H}(\omega) = -\frac{R_f}{R_i + 1/j\omega C_i} = -\frac{j\omega C_i R_f}{1 + j\omega C_i R_i}$$

$$\omega \rightarrow 0, |\mathbf{H}(\omega)| \rightarrow 0 \quad \omega \rightarrow \infty, |\mathbf{H}(\omega)| \rightarrow -\frac{R_f}{R_i}$$

$$\omega_c = \frac{1}{R_i C_i}$$

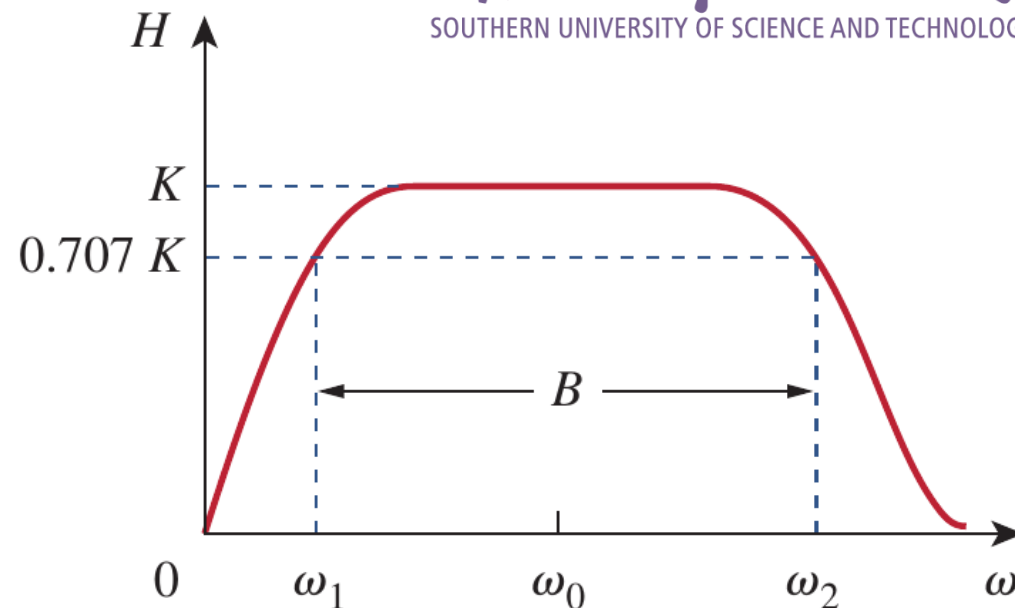
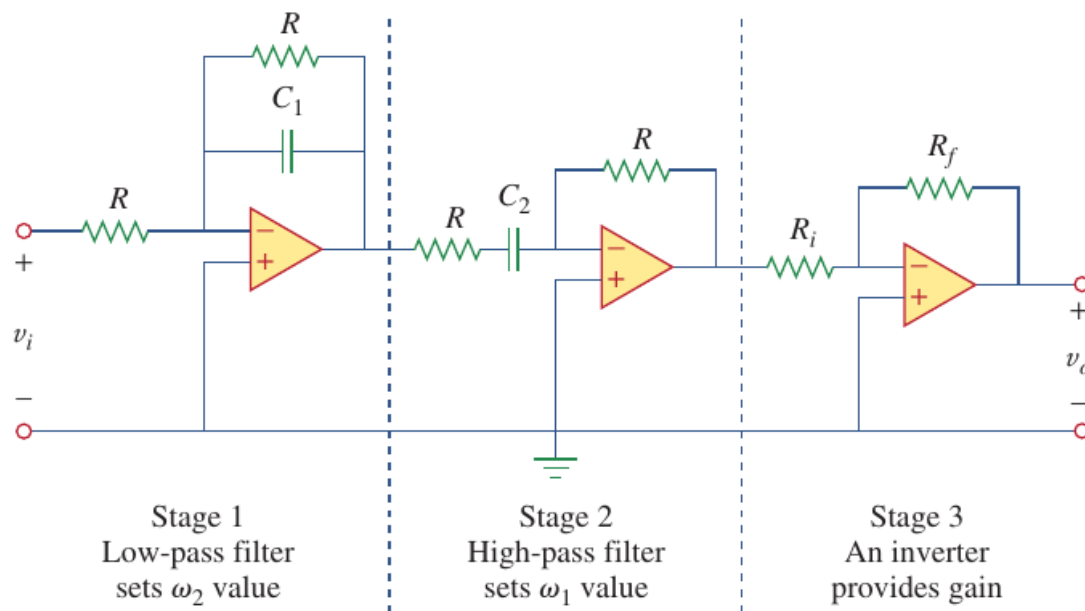
11.10 Active filters (e)

- Active bandpass filters:



11.10 Active filters (f)

- Active bandpass filters:



$$\omega_1 = \frac{1}{RC_2} = \frac{1}{R_{high}C_{high}}, \quad \omega_2 = \frac{1}{RC_1} = \frac{1}{R_{low}C_{low}}$$

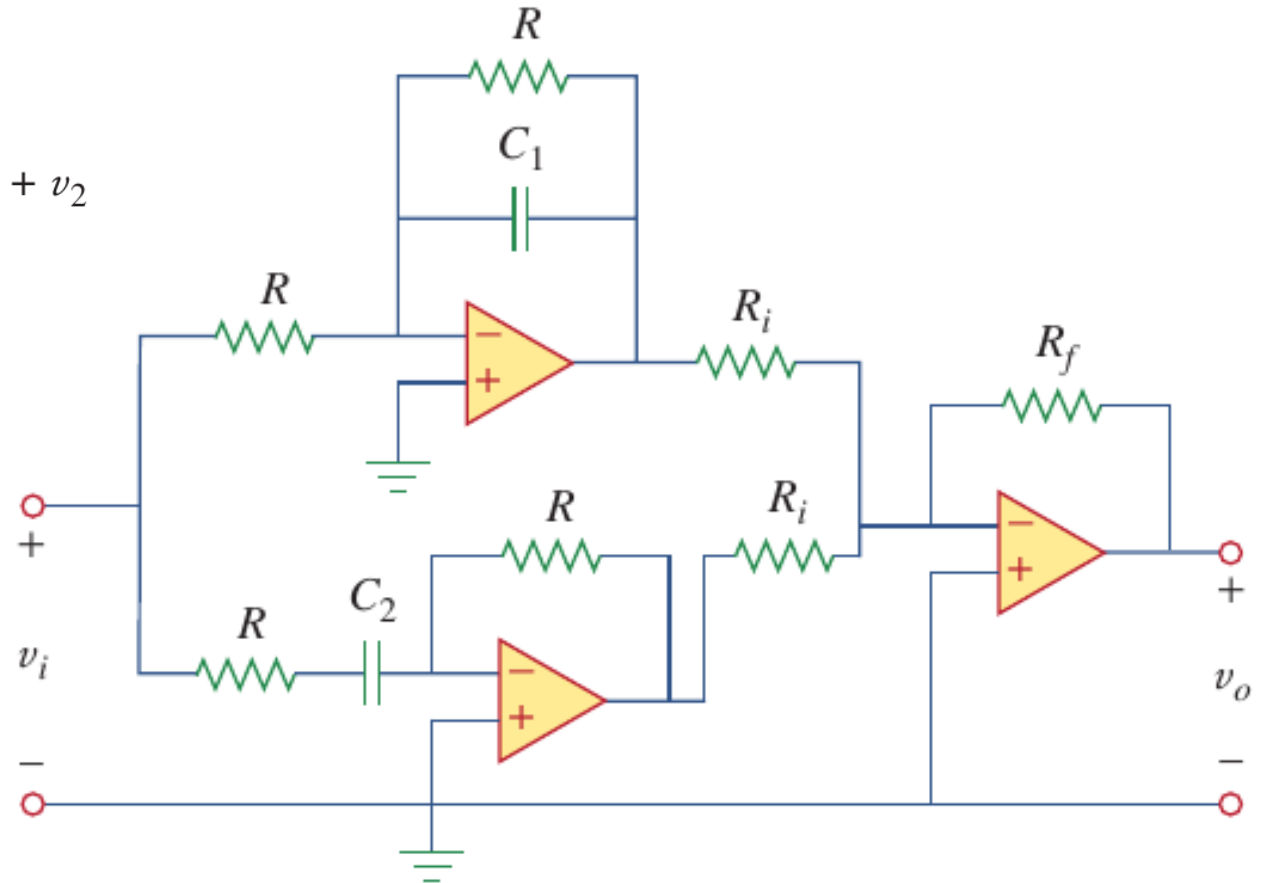
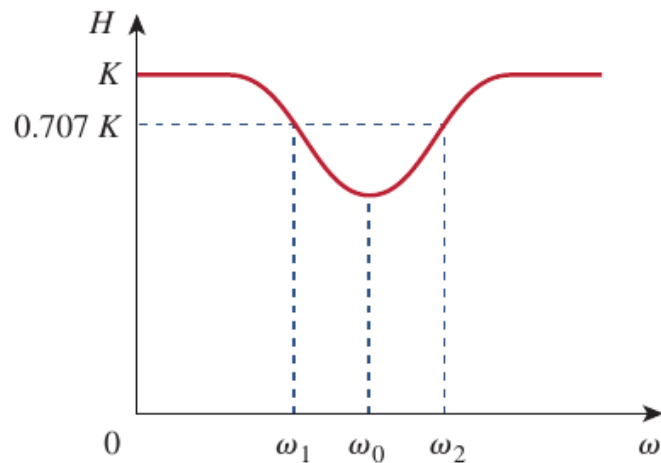
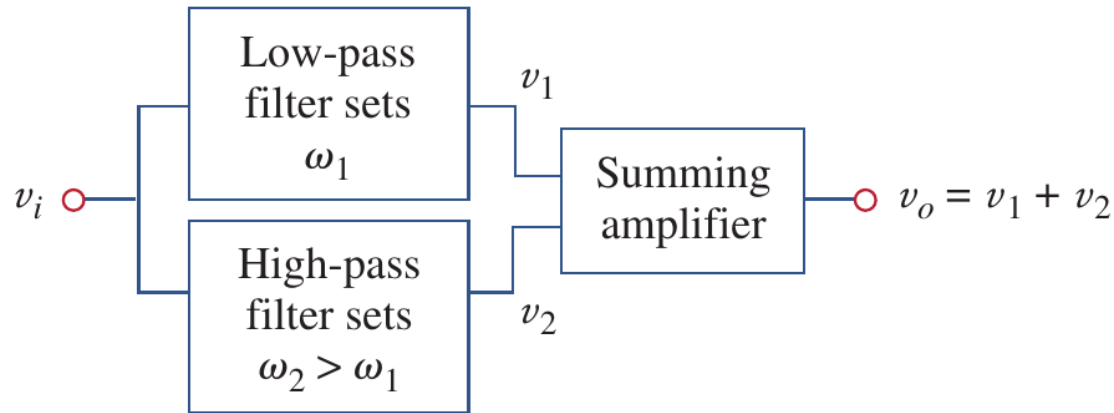
In the pass bands:

$$K = \frac{R_f}{R_i} \frac{\omega_2}{\omega_1 + \omega_2}$$

$$\begin{aligned} \mathbf{H}(\omega) &= \frac{\mathbf{V}_o}{\mathbf{V}_i} = \left(-\frac{1}{1 + j\omega C_1 R} \right) \left(-\frac{j\omega C_2 R}{1 + j\omega C_2 R} \right) \left(-\frac{R_f}{R_i} \right) \\ &= -\frac{R_f}{R_i} \frac{1}{1 + j\omega C_1 R} \frac{j\omega C_2 R}{1 + j\omega C_2 R} \end{aligned}$$

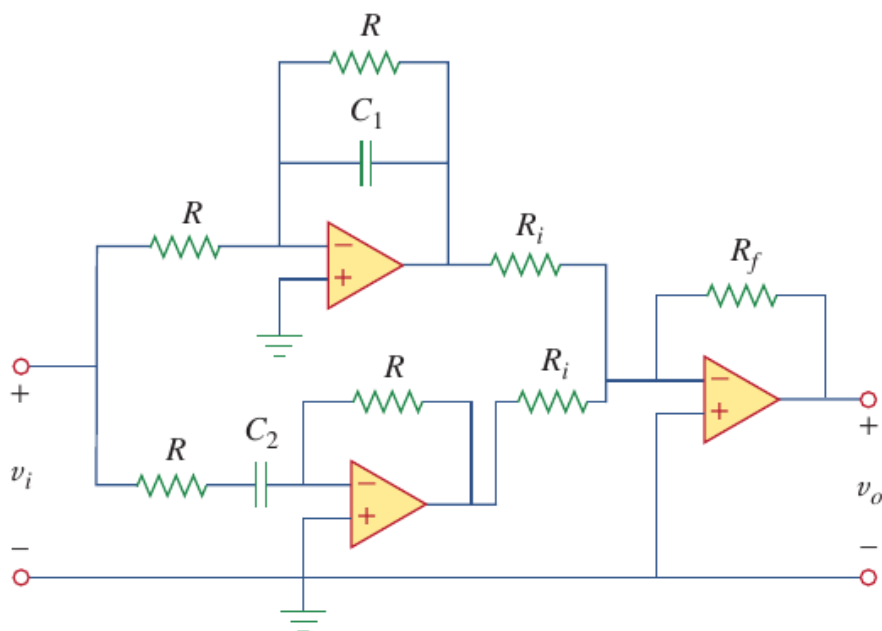
11.10 Active filters (g)

- Active bandreject filters (or Notch filters):



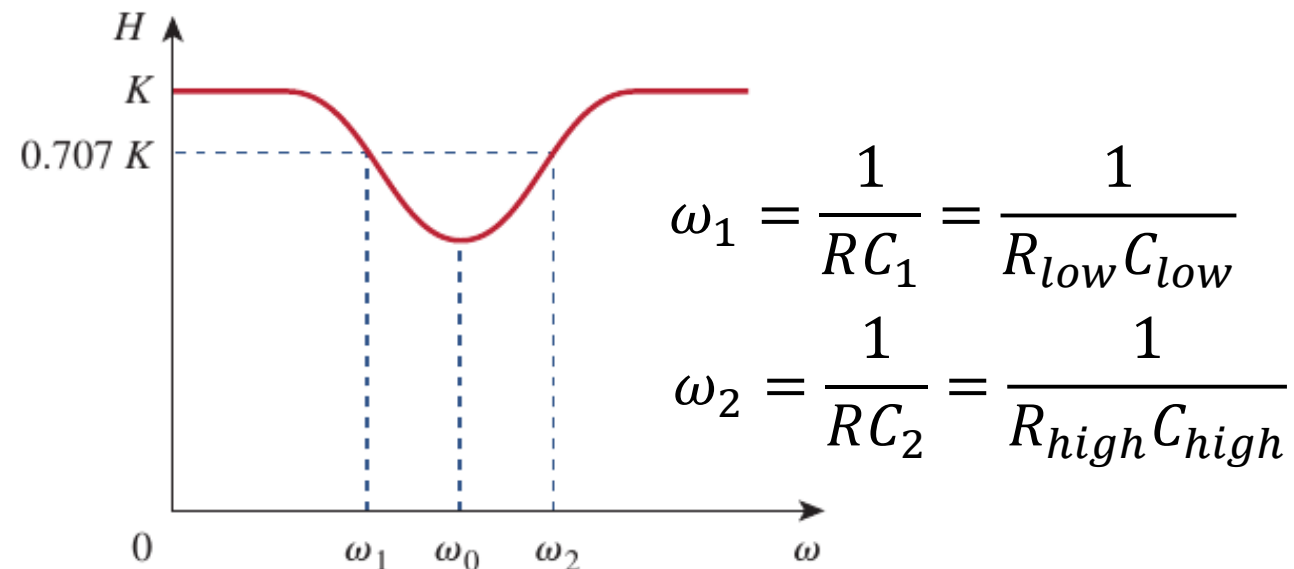
11.10 Active filters (h)

- Active bandreject filters:



$$\mathbf{H}(\omega) = \frac{R_f}{R_i} \left(\frac{1}{1 + j\omega/\omega_2} + \frac{j\omega/\omega_1}{1 + j\omega/\omega_1} \right)$$

$$= \frac{R_f}{R_i} \frac{(1 + j2\omega/\omega_1 + (j\omega)^2/\omega_1\omega_2)}{(1 + j\omega/\omega_2)(1 + j\omega/\omega_1)}$$



In the pass bands: $K = \frac{R_f}{R_i}$

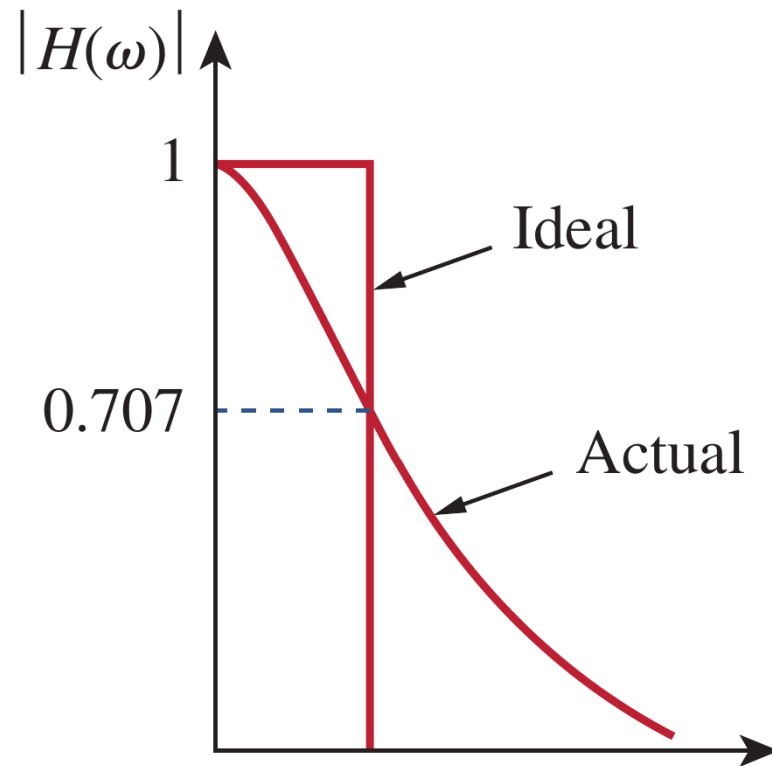
At ω_c : $H(\omega_0) = \frac{R_f}{R_i} \frac{2\omega_1}{\omega_1 + \omega_2}$

Lesson 11 Frequency response

11.1-11.4	Transfer function, The Decibel Scale, Bode Plots	Slides 09-20
11.5-11.6	Lowpass and highpass filters	Slides 22-37
11.7-11.8	Series and parallel resonances	Slides 39-54
11.9	Bandpass and bandstop filters	Slides 56-66
11.10	Active filters	Slides 68-75
11.11	High-order filters and scaling	Slides 77-81
11.12	Summary	Slides 83-85

11.11 High-order filters and Scaling (a)

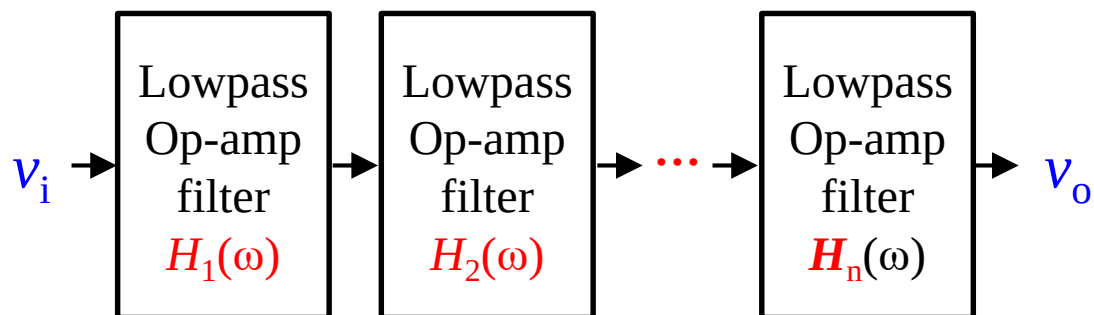
- How to make the filters more ideal?



- You have probably noticed that all of the filter circuits we have examined so far, both passive and active, are nonideal.
- Remember that an ideal filter has a discontinuity at the point of cutoff, which sharply divides the passband and the stopband.
- Although we cannot hope to construct a circuit with a discontinuous frequency response, we can construct circuits with a sharper, yet still continuous, transition at the cutoff frequency.

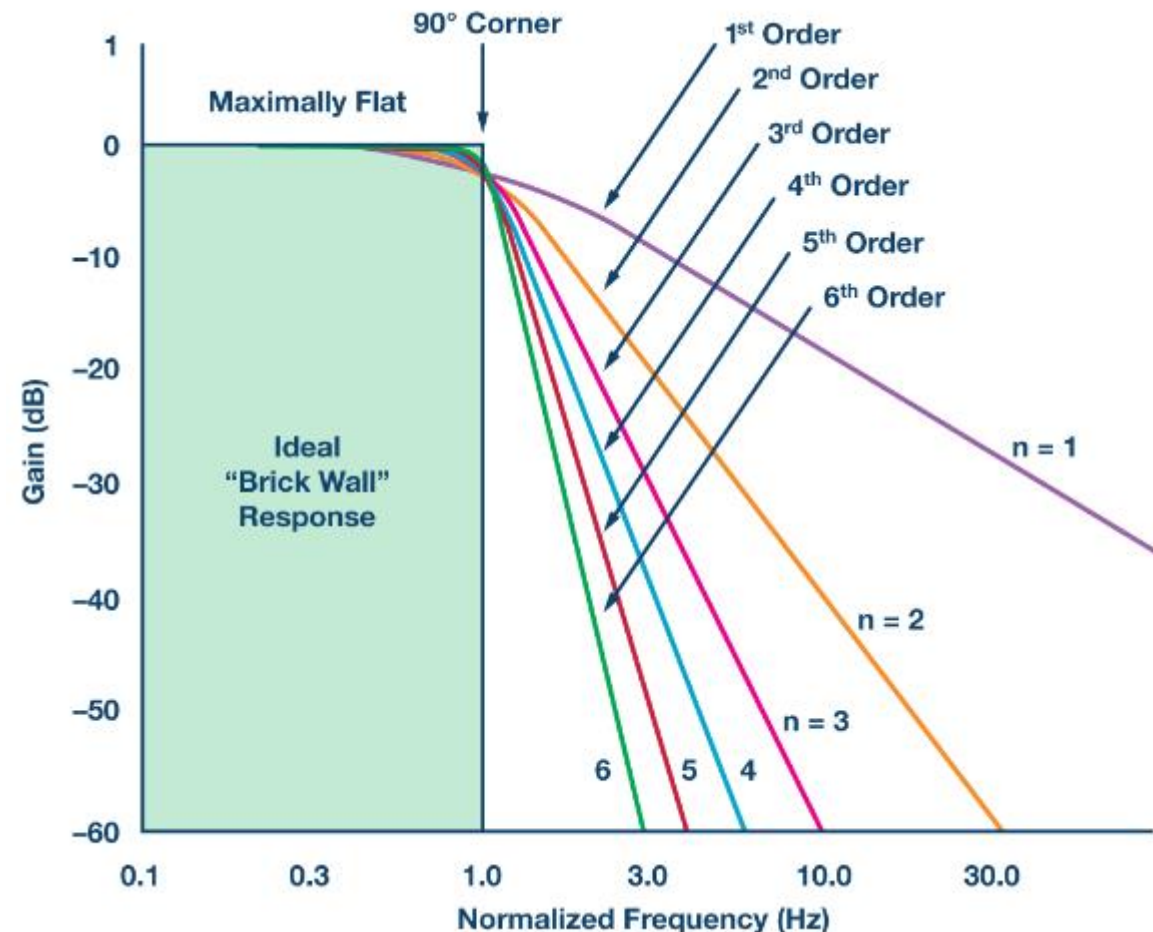
11.11 High-order filters and Scaling (b)

- High-order filters:



$$H_{total}(\omega) = H_1 H_2 \cdots H_n$$

- a) High-order filters result in a sharper, yet still continuous transition at ω_c .
- b) The ω_c is changing.
- c) If these active filters are identical, we can use scaling (比例变换) to analyze them.



11.11 High-order filters and Scaling (c)

- **Scaling:**

- a) In designing a circuit (e.g. filters), it is convenient to work with normalized values for elements and then transform them to realistic ones. This transforming process is called scaling, involving magnitude scaling and frequency scaling.

Magnitude scaling

Increasing all Z in the network by a factor (K_m) while keeping the frequency response unchanged.

$$\begin{aligned} R' &= K_m R, & L' &= K_m L \\ C' &= \frac{C}{K_m}, & \omega' &= \omega \end{aligned}$$

Frequency scaling

Shifting the frequency response of a network while leaving the element impedance the same.

$$\begin{aligned} R' &= R, & L' &= \frac{L}{K_f} \\ C' &= \frac{C}{K_f}, & \omega' &= K_f \omega \end{aligned}$$

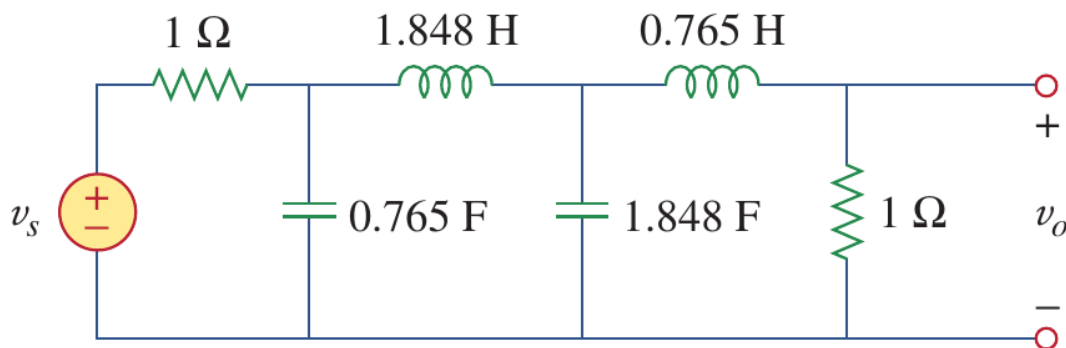
Magnitude & Frequency scaling

$$\begin{aligned} R' &= K_m R, & L' &= \frac{K_m}{K_f} L \\ C' &= \frac{1}{K_m K_f} C, & \omega' &= K_f \omega \end{aligned}$$

11.11 High-order filters and Scaling (d)

- Example 11.06:

Example 11.06: The following circuit is a fourth-order Butterworth lowpass filter. The filter is designed such that the cutoff frequency $\omega_c = 1$ rad/s. If we scale the circuit for a cutoff frequency of 50 kHz using 10 k Ω resistors, find the new values of C and L .



Solution:

If the cutoff frequency is to shift from $\omega_c = 1$ rad/s to $\omega'_c = 2\pi(50)$ krad/s, then the frequency scale factor is

$$K_f = \frac{\omega'_c}{\omega_c} = \frac{100\pi \times 10^3}{1} = \pi \times 10^5$$

Also, if each $1\text{-}\Omega$ resistor is to be replaced by a $10\text{-k}\Omega$ resistor, then the magnitude scale factor must be

$$K_m = \frac{R'}{R} = \frac{10 \times 10^3}{1} = 10^4$$

Using Eq. (14.86),

$$L'_1 = \frac{K_m}{K_f} L_1 = \frac{10^4}{\pi \times 10^5} (1.848) = 58.82\ \text{mH}$$

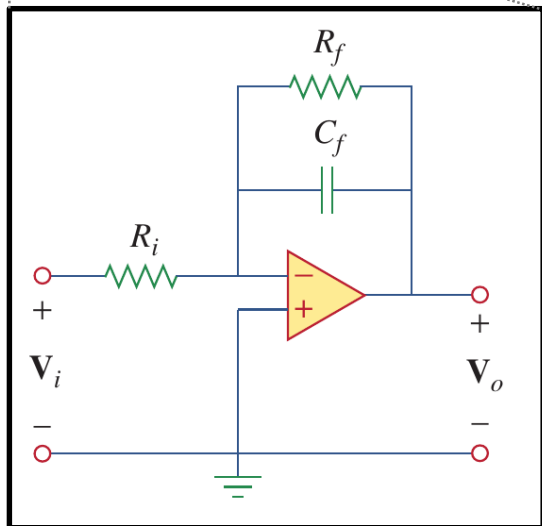
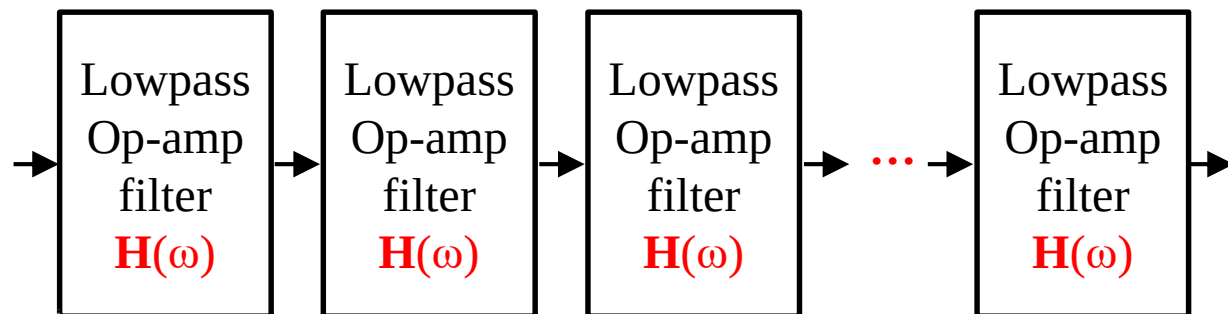
$$L'_2 = \frac{K_m}{K_f} L_2 = \frac{10^4}{\pi \times 10^5} (0.765) = 24.35\ \text{mH}$$

$$C'_1 = \frac{C_1}{K_m K_f} = \frac{0.765}{\pi \times 10^9} = 243.5\ \text{pF}$$

$$C'_2 = \frac{C_2}{K_m K_f} = \frac{1.848}{\pi \times 10^9} = 588.2\ \text{pF}$$

11.11 High-order filters and Scaling (e)

- High-order active filters :



Identical lowpass filters.

For each filter:

$$|H(\omega)| = \left(\frac{R_f}{R_i}\right)^2 \frac{\omega_c^2}{\omega^2 + \omega_c^2}$$

For n stages:

$$|H_{\text{eff}}(\omega)| = \left(\frac{R_f}{R_i}\right)^{2n} \left(\frac{\omega_c^2}{\omega^2 + \omega_c^2}\right)^n$$

At the effective cutoff frequency $\omega_{c,\text{eff}}$:

$$|H_{\text{eff}}(\omega_{\text{eff}})| = \frac{1}{\sqrt{2}} H_{\text{eff,max}} = \frac{1}{\sqrt{2}} \left(\frac{R_f}{R_i}\right)^{2n}$$

$$\Rightarrow \left(\frac{\omega_c^2}{\omega_{c,\text{eff}}^2 + \omega_c^2}\right)^n = \frac{1}{\sqrt{2}}$$

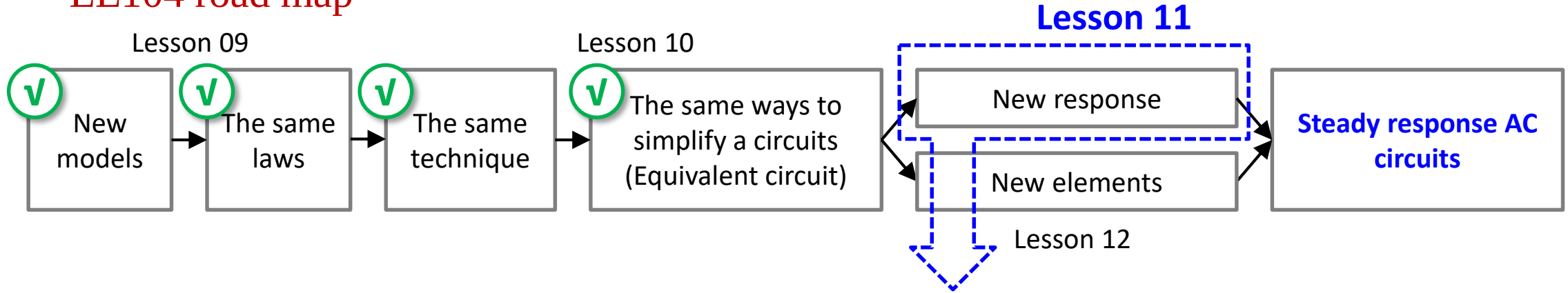
$$\Rightarrow \omega_{c,\text{eff}} = \omega_c \sqrt[n]{\sqrt{2} - 1}$$

Lesson 11 Frequency response

11.1-11.4	Transfer function, The Decibel Scale, Bode Plots	Slides 09-20
11.5-11.6	Lowpass and highpass filters	Slides 22-37
11.7-11.8	Series and parallel resonances	Slides 39-54
11.9	Bandpass and bandstop filters	Slides 56-66
11.10	Active filters	Slides 68-75
11.11	High-order filters and scaling	Slides 77-81
11.12	Summary	Slides 83-85

11.12 Summary (a)

- EE104 road map



Frequency response

- Transfer function
- Passive filters: lowpass and highpass filters
- Resonance + bandpass and bandstop filters
- Active filters
- High-order filter

11.12 Summary (a)

- EE104 road map

Problems

14.29, 14.46, 14.57, 14.59, 14.66

- (a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.
- (b) There are plenty of problems in the textbook, try to complete as many as you can.

Polar Stratospheric Clouds

